eLife

Neuroscience

Astrocytes gate long-term potentiation in hippocampal interneurons

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint posted
September 27, 2023 (this version)

Posted to bioRxiv
June 24, 2023

Sent for peer review
June 20, 2023

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Abstract

Long-term potentiation is involved in physiological process like learning and memory, motor learning and sensory processing, and pathological conditions such as addiction. In contrast to the extensive studies on the mechanism of long-term potentiation on excitatory glutamatergic synapse onto excitatory neurons ($LTP_{E\rightarrow E}$), the mechanism of LTP on excitatory glutamatergic synapse onto inhibitory neurons ($LTP_{E\rightarrow I}$) remains largely unknown. In the central nervous system, astrocytes play an important role in regulating synaptic activity and participate in the process of $LTP_{E\rightarrow E}$, but their functions in $LTP_{E\rightarrow I}$ remain incompletely defined. Using electrophysiological, pharmacological, confocal calcium imaging, chemogenetics and behavior tests, we studied the role of astrocytes in regulating $LTP_{E\rightarrow I}$ in the hippocampal CA1 region and their impact on cognitive function. We show that $LTP_{E\rightarrow I}$ in stratum oriens of hippocampal CA1 is astrocyte independent. However, in the stratum radiatum, synaptically released endocannabinoids increases astrocyte Ca^{2+} via type-1 cannabinoid receptors, stimulates D-serine release, and potentiate excitatory synaptic transmission on inhibitory neuron through the activation of (N-methyl-D-aspartate) NMDA receptors. We also revealed that chemogentic activation of astrocytes is sufficient to induce NMDA-dependent *de novo* $LTP_{E\rightarrow I}$ in the stratum radiatum of hippocampus. Furthermore, we found that disrupt $LTP_{E\rightarrow I}$ by knockdwon γCaMKII in interneurons of stratum radiatum resulted in dramatic memory impairment. Our findings suggest that astrocytes release D-serine, which activates NMDA receptors to regulate $LTP_{E\rightarrow I}$, and that cognitive function is intricately linked with the proper functioning of this $LTP_{E\rightarrow I}$ pathway.

eLife assessment

The study presents **valuable** insights into the regulation of astrocytes in the long-term potentiation of excitatory synapses onto inhibitory interneurons. However, reviewers identified concerns regarding originality and proper acknowledgment of replicated work, representation of interneuron diversity, and the robustness of certain conclusions. The strength of evidence provided is deemed **incomplete**, necessitating significant revisions for clarity, and accuracy, and to address highlighted concerns.

Materials and Methods

Animals

Our study was conducted in accordance with the Guide for the Care and Use of Laboratory Animals and was approved by the ethics committee of the Hangzhou City University (registration number: 22061). C57BL/6 male mice (2-4 months) were purchased from Hangzhou Ziyuan Laboratory Animal Corporation and housed in groups of three to four per cage. The mice were maintained on a 12-hour light/dark cycle and were provided with *ad libitum* access to food and water.

Stereotactic virus injection

Stereotactic virus injection was conducted as described previously (1, 2). Briefly, adult mice were deeply anesthetized with sodium pentobarbital (50 mg/kg) and secured in a stereotaxic device with ear bars (RWD, 68930) while their body temperature was maintained at approximately 37 °C using a heating blanket. Their hair was removed using a razor and the skin was sterilized with iodophor. A 1-cm incision in the midline using sterile scissors. Small burr holes were drilled bilaterally using an electric hand drill at the following coordinates: anteroposterior (AP), 2.3 mm from Bregma; mediolateral (ML), ±1.4 mm. Virus particles were then injected bilaterally into the stratum radiatum (1.2 mm from the pial surface) using glass pipettes connected to an injection pump (RWD, R480). The injection rate was controlled at 1 nl/s using the pump. To allow the virus disseminate into the tissue, glass pipette was left in place for 10 minutes after each injection. After the injection, the pipettes were gradually removed and the wound was sutured up. For hippocampal interneuron physiological recording, 200 nl AAV2/9 mDLx EGFP (6.4×10^{11} gc/ml) were injected. For hippocampal slice Ca^{2+} imaging, 500 nl AAV2/5 GfaABC$_1$D GCaMP6f (1.2×10^{12} gc/ml) with injected alone or mixed with 500 nl AAV2/5 GfaABC$_1$D hM3D (Gq) mCherry (5.3×10^{12} gc/ml). To knockdown γCaMKII in hippocampal interneuron, an shRNA sequence (5'-GCAGCTTGCATCGCCTATATC-3') was used. A total of 200 nl of AAV2/9 mDLx γCaMKII shRNA (6.4×10^{12} gc/ml) were injected. All virus were generated by Brainvta and Sunbio Medical Biotechnology (Wuhan, *https://www.brainvta.tech/* and Shanghai, *http://www.sbo-bio.com.cn/*).

Electrophysiology

Mice were anesthetized with isoflurane, and their brains were quickly extracted and immersed in an ice-cold solution which containing (in mM) 235 Sucrose, 1.25 NaH_2PO_4, 2.5 KCl, 0.5 $CaCl_2$, 7 $MgCl_2$, 20 Glucose, 26 $NaHCO_3$, and 5 Pyruvate (pH 7.3, 310 mOsm, saturated with 95% O_2 and 5% CO_2). Coronal hippocampus slices (300-350 μm) were prepared with a vibrating slicer (Leica, V T1200) and incubated for 30 - 40 min in artificial cerebrospinal fluid (ACSF) containing (in mM) 26 $NaHCO_3$, 2.5 KCl, 126 NaCl, 20 D-glucose, 1 sodium pyruvate, 1.25 NaH_2PO_4, 2 $CaCl_2$ and 1 $MgCl_2$ (pH 7.4, 310 mOsm, saturated with 95% O_2 and 5% CO_2).

The slices were transferred to an immersed chamber and continuously perfused with oxygen-saturated ACSF and GABA receptor blockers, picrotoxin (100 μM) and CGP55845 (5 μM), at a rate of 3 ml/min. Interneurons in the stratum radiatum or stratum oriens were visualized using infrared differential interference contrast and epifluorescence imaging. Perforated-path recordings were conducted as previously described (3). Briefly, perforated whole-cell recording from stratum radiatum or stratum oriens interneurons were made with pipettes filled with solution containing (in mM) 136 K-gluconate, 9 NaCl, 17.5 KCl, 1 $MgCl_2$, 10 HEPES, 0.2 EGTA, 25 μM Alexa 488, amphotericin B (0.5 mg ml^{-1}) and small amounts of glass beads (5–15 μm in diameter; Polysciences, Inc., Warminster, PA, USA) (pH 7.3, 290 mOsm). The pathched neuron was intermittently imaged with epifluorescence to monitor dye penetration. If the patch ruptured spontaneously, the experiment was discontinued.

Whole-cell voltage-clamp recordings were made from either stratum radiatum or stratum oriens interneurons using pipettes with resistance of 3-4 MΩ and filled with a solution containing (in mM) 4 ATP-Na_2, 0.4 GTP-Na, 125 $CsMeSO_3$, 10 EGTA, 10 HEPES, 5 4-AP, 8 TEA-Cl, 1 $MgCl_2$, 1 $CaCl_2$ (pH 7.3–7.4, 280–290 mOsm).

Whole-cell current-clamp recording were made from stratum interneurons using pipettes with resistance of 4 - 6 MΩ and filled with a solution containing (in mM) 125 K-Gluconate, 2 $MgCl_2$, 10 HEPES, 0.4 Na^+-GTP, 4 ATP-Na_2, 10 Phosphocreatine disodium salt, 10 KCl, 0.5 EGTA (pH 7.3–7.4, 280–290 mOsm).

Whole-cell recording were performed on stratum radiatum astrocytes using pipettes with resistance of 8 - 10 MΩ and filled with an intracellular solution containing (in mM) 130 K-Gluconate, 20 HEPES, 3 ATP-Na_2, 10 D-Glucose, 1 $MgCl_2$, 0.2 EGTA (pH 7.3-7.4, 280-290 mOsm). In a subset of experiments, 0.14 mM $CaCl_2$ and 0.45 mM EGTA were included in the upper intracellular solution to maintain a stable level of astrocytic concentration (calculation by Web-MaxChelator) (2, 4).

Electrical stimuli was delivered via theta glass pipettes in the Schaffer Collateral of stratum radiatum, with a 30 s inter-trial interval during baseline and after LTP induction. Evoked EPSPs were recorded in current-clamp model at the resting membrane potential of the stratum radiatum interneuron. After a 10-min stable baseline period, LTP was induced by applying theta-burst stimulation [TBS; five bursts at 200- ms intervals (5 Hz), each burst consisting of five pulses at 100 Hz], paired with five 60- ms long depolarizing steps to −10 mV. Six episodes of TBS paired with depolarization were given at 20-s intervals. In stratum oriens, LTP was induced by applying TBS paired with five 60-ms long hyperpolarizing steps to −90 mV.

sEPSCs were recorded in whole-cell model at −70 mV in the presence of picrotoxin (100 μM) and D-AP5 (50 μM). Paired pulses were delivered at an inter-pulse intervals of 50 ms, and the paired-pulse ratio was calculated by dividing the peak amplitude of the second EPSC by the peak amplitude of first EPSC. To analyze action potential properties, interneurons were recorded at rest and depolarized with 500-ms current injection pulses at 10-pA increments.

To avoid NMDAR-mediated signaling rapidly washing out when recording interneuron in whole-cell mode, electrical stimuli was delivered with a 5 s inter-trial interval to measure the I-V relationship for NMDAR-mediated EPSPs within 5 minutes of breaking in. The NMDA/AMPA ratio in the stratum radiatum or stratum oriens interneurons was calculated by measuring the amplitude of NMDAR-mediated EPSCs at + 60 mV (50 ms after stimulation) and the peak amplitude of AMPAR-mediated EPSC recorded at −60 mV. To measure pure NMDAR-mediated EPSCs in interneurons at resting membrane potential in perforated whole-cell recording model, Mg^{2+} free ACSF was used, and D-AP5 (50 μM) and picrotoxin (100 μM) were present in ACSF.

Axopatch 700B amplifiers were utilized for patch-clamp recordings (Molecular Devices). The data was filtered at 6 kHz and sampled at 20 kHz before being processed off-line using the pClampfit 10.6 program (Molecular Devices). Negative pules (−10 mV) were employed to monitor series resistances and membrane resistances. The data was included to analysis if the series resistances varied by less than 20% over the course of the trial. All experiments were carried out at 32°C.

Behavioral assays

For the contextual and cued fear conditioning test, the mouse was habituated to the environment and handled for three consecutive days. On the fourth day, the mice were allowed to explore the conditioning cage for 2 min, after which they received three moderate tone-shock pairs [30 stone (80 dB, 4 kHz) co-terminating with a foot shock (0.4 mA, 2 s)]. Following conditioning, the mice were returned to their home cages. The next day, they were place in the same conditioning cage but without receiving any foot shocks, and their freezing behavior was analyzed for the first 5 min. Four hours after the contextual fear conditioning test, the mice were placed in a novel cage with a different shape and texture of the floors compared to the conditioning cage). They were allowed to freely explore the new environment for 2 min and then subjected to 3 tones stimulations [(80 dB, 4 kHz) lasting for 30 s] separated by intervals of 90s, but without receiving any foot-shocks. Once the final tone had ended, all of the mice were allowed to freely explore the chamber for an additional 90 s. Fear responses were measured by calculating the freezing values of the mice using Packwin software (Panlab, Harvard Apparatus, USA). All apparatus were carefully cleaned with 30% ethanol in between tests.

Immunohistochemistry

Immunohistochemistry was conducted as described previously (1, 5, 6). Briefly, mice were administered a single intraperitoneal injection of 50 mg/kg sodium pentobarbital for anesthesia and were transcardially perfused with phosphate-buffered saline (PBS). After the liver and lungs had become bloodless, the mice were perfused with 4% paraformaldehyde (PFA) in 0.1 M PBS. The brains were quickly removed and placed in 4% PFA at 4°C overnight. Following this, the tissues were cryoprotected in successive concentrations of 10%, 20%, and 30% sucrose before being sliced into 20 μm sections using a freezing microtome. After being washed multiple times with PBS, the sections were blocked for 1.5 hours at room temperature (22-24°C) in a blocking solution consisting of 5% Bovine Serum Albumin (BSA) and 1% Triton X-100. Following the blocking step, the sections were incubated with primary antibodies overnight at 4 °C. Antibodies used were as follows: mouse monoclonal anti-GFAP (1/1000, Cell Signaling Technology Cat #3670, RRID:AB_561049), mouse monoclonal anti-NeuN (1/500, Millipore Cat# MAB377, RRID: AB_2298772), mouse monoclonal GAD67 (1/500, Synaptic Systems Cat# 198 006, RRID: AB_2713980). After wash several times in PBS, sections were then incubated with the following secondary antibody: Alexa Fluor 594 goat anti-mouse (1/1000, Cell Signaling Technology Cat# 8890, RRID: AB_2714182) at room temperature for 2 h. Afterwards, the sections were rinsed several times in PBS and incubated with DAPI for 5 minutes at room temperature. Following this, the sections were rinsed again and mounted with Vectashield mounting medium. Images were examined using a confocal laser scanning microscope (Olympus, VT1000) and analyzed using ImageJ (NIH, RRID: SCR_003070).

Ca^{2+} imaging

Ca^{2+} signals in hippocampal astrocytes were observed under a confocal microscopy (Fluoview 1000; Olympus) with a 40x water immersion objective lens (NA = 0.8). GCaMP6f was excited at 470 nm. Astrocytes located in the hippocampal CA1 region and at least 40 μm away from the slice surface were selected for imaging. Images were acquired at 1 frame per 629 ms. Hippocampal slices were maintained in ACSF containing picrotoxin (100 μM), and CGP55845 (5 μM) using a perfusion system. A single theta burst stimulation (TBS) was used to stimulate the neuron. CNO (5 μM) was bath-applied to activate hM3D(Gq)-mediated Ca^{2+} signals. The Ca^{2+} signal analysis was described previously (1, 2, 5, 6). In some experiments, image stacks comprising 10-15 optical sections with 1 μM z-spacing were acquired to facilitate the identification of GCaMP6f and hM3Dq (mCherry) co-expression in astrocytes. Briefly, movies were registered using the StackReg plugin of ImageJ to eliminate any x-y drift. Ca^{2+} signals were then analyzed in selected ROIs using the Time Series Analyzer V3 plugin of ImageJ. GCaMP6f fluorescence was calculated as $\Delta F/F = (F-F_0)/F_0$. Mean ΔF/F of Ca^{2+}signals were analyzed using Clampfit 10.6.

Statistics

All data processing, figure generation, layout, and statistical analysis were performed using Clampfit 10.6, Prism, MATALAB and Coreldraw. To assess the normality of the data, the Shapiro-Wilk test was used. If the results of the test were not statistically significant ($p> 0.05$), the data were assumed to follow a normal distribution, and a paired t-test was employed. On the other hand, if the test was statistically significant, a Wilcoxon Signed Rank Test or Mann-Whitney Ran Sum Test was utilized. To statistically analyze cumulative frequency distributions, the Kolmogorov-Smirnov test was employed. When comparing two groups, either a Wilcoxon signed-rank test or a Student's t-test (paired or unpaired) was used. When comparing three groups, One-way repeated measures (RM) ANOVA followed by Dunnett's post-hoc test was used.

All values were presented as mean ± standard error of the mean (SEM). Data were considered significantly different when the p value was less than 0.05 (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$). The numbers of cells or mice used in each analysis were indicated in figure legends.

Introduction

Long-term potentiation (LTP) was originally discovered as a long-term increase in synaptic efficacy at glutamatergic synapses onto granule cells of the hippocampus (7). This form of plasticity has been well documented at CA1 pyramidal cells, where synaptic plasticity can be induced by the activation of postsynaptic (N-methyl-D-aspartate) NMDA receptors (NMDARs), voltage-dependent Ca^{2+} channels, and metabotropic glutamate receptors (mGluRs) (8, 9) . Numerous studies have provided compelling evidence that LTP in hippocampal network occurs not only at excitatory glutamatergic synapse onto excitatory pyramidal and granule cells ($LTP_{E\rightarrow E}$), but also at excitatory glutamatergic synapses onto inhibitory interneurons ($LTP_{E\rightarrow I}$) (10–14). However, the mechanism behind $LTP_{E\rightarrow I}$ remains controversial due to the heterogeneous expression of proteins in various types of hippocampal interneurons and the absence of synaptic spines between excitatory inputs on interneurons (10, 15). Two distinct forms of $LTP_{E\rightarrow I}$, namely NMDARs-dependent LTP and NMDARs- independent $LTP_{E\rightarrow I}$, have been observed in hippocampal interneurons.

Astrocytes are the most prevalent glial cell type in the central nervous system (CNS), and play a critical role in regulating the development and function of the nervous system (16, 17). Astrocyte display multipolar branches with numerous micro- processes that allow them to closely associate with blood vessels, neuronal cell bodies and axons, other glial cells and as well as synapses (18–20). They also express varies ion channels, transporters and neurotransmitter receptors (21, 22). With these membrane proteins, they can sense neuronal activity and exhibit increases in intracellular Ca^{2+} in reaction to neurotransmitters and, in turn, they release neuroactive chemicals called gliotransmitters that regulate synaptic transmission and plasticity (17, 23–25). Numerous studies have been shown that the release of D-serine, a co-agonist of NMDAR, from astrocytes is capable of enabling LTP in cultures, in slice and as well as in vivo (4, 26–29). Given the growing number of studies demonstrating the direct roles that astrocytes play in regulating $LTP_{E\rightarrow E}$, it is of particular interest to understand whether and how astrocytes in modulation LTP of excitatory postsynaptic currents in interneuron.

In this particular study, our focus was on the interneurons distributed in the stratum radiatum layer of the CA1 region of the hippocampus. Approximately 80% of these interneurons exhibit NMDAR-dependent $LTP_{E\rightarrow I}$ when presynaptic stimulation is paired with postsynaptic depolarization. We found that blocking astrocyte metabolism and clamping of astrocyte Ca^{2+} signaling can prevent the $LTP_{E\rightarrow I}$ in large NMDAR- containing interneurons, which could be rescued by bath application of D-serine. Furthermore, pharmacological and Ca^{2+} imaging studies have shown that astrocytes respond to Schaffer Collateral stimulation with Ca^{2+} increases through activation of type-1 cannabinoid receptors (CB1R), which stimulates the release of D-serine, and further regulates $LTP_{E\rightarrow I}$ through binding to the glycine site of NMDARs. We also found that activating astrocytes with Gq designer receptors exclusively activated by designer drugs (DREADDs) induced a potentiation of excitatory to inhibitory synapses in stratum radiatum. Additionally, knockdown γCaMKII hampered $LTP_{E\rightarrow I}$ in stratum radiatum of hippocampus in brain slice and disrupted contextual fear conditioning memory in vivo. Taken together, these results are the first to indicate that astrocytes are an integral component of a form of long-term synaptic plasticity between glutamatergic neurons and GABAergic interneurons, and that memory is also regulated by $LTP_{E\rightarrow I}$.

Results

Astrocyte play a role in the formation of NMDAR-dependent $LTP_{E\rightarrow I}$ in CA1 stratum radiatum

To visualize CA1 stratum radiatum interneurons, we delivered an adeno-associated virus serotype 2/9 (AAV2/9) vector encoding EGFP under the control of the interneuronal mDLx promoter (AAV2/9-mDLx-EGFP) to this region (30). Within the virally transduced region, EGFP expression was limited to the interneuron, with high penetrance (>98% of the GAD67 cells expressed EGFP) **(Fig. S1 A-C)** and almost complete specificity (>98% EGFP positive cells were also GAD67 positive) **(Fig. S1 D)**. These immunostaining results indicate that EGFP expression was limited to the interneuron in the CA1 region of stratum radiatum.

We next examined whether expressing EGFP in interneurons affects their membrane properties and synaptic transmission. To examine these, we performed whole-cell patch recordings of $EGFP^+$ interneurons and putative interneurons in the CA1 stratum radiatum of the hippocampus in the presence of GABAA receptor blocker picrotoxin. We found that the excitability and resting membrane potential are no difference between these $EGFP^+$ interneurons and putative interneurons **(Fig. S2 A-C)**. Moreover, we also found that no difference in spontaneous excitatory postsynaptic currents (sEPSC) and paired-pulse ratio (PPR) at 50 ms interpulse intervals between $EGFP^+$ interneurons and putative interneurons **(Fig. S2 D-H)**. These results indicate that AAV injection and expression exogenous protein in interneurons has no effect on the membrane properties and baseline synaptic transmission of interneuron.

To avoid some necessary ingredient for $LTP_{E\rightarrow I}$ induction be diluted from the cytoplasm, perforate patch-clamp were used to record EPSPs from CA1 stratum radiatum interneurons. After a 10 min baseline recording, we delivered theta burst stimulation (TBS) consisting of 100 Hz stimulation with 25 pulses, delivered in six trains separated by 20-second intervals. The TBS was applied to induce $LTP_{E\rightarrow I}$ of the excitatory inputs to CA1 interneurons. The interneurons were depolarized to −10 mV using a voltage-clamp model during the TBS delivery. Under this condition, we found that in 8 out of 10 cells were induced $LTP_{E\rightarrow I}$ lasting at least 45 min **(Fig. 1 A-C)**. We repatched 6 of these successful to induction $LTP_{E\rightarrow I}$ cells and randomly patched 2 $EGFP^+$ in stratum radiatum in the whole-cell voltage-clamp model. The results indicated that all cells showed a linear current-voltage (I-V) curve for AMPAR, as well as a significant component of NMDAR-mediated currents

(Fig. S3). In addition, we found that the induction of $LTP_{E\rightarrow I}$ was completely blocked by NMDAR blocker D- AP5 **(Fig. 1 A-C)**. This result confirmed that $LTP_{E\rightarrow I}$ in CA1 stratum radiatum interneurons is NMDAR dependent (31, 32).

To investigate whether astrocytes were involved in NMDAR-dependent $LTP_{E\rightarrow I}$ in CA1 stratum radiatum interneurons, we treated slice with fluoroacetate (FAC, 5 mM) to block glial metabolism (4, 33). The results showed that the induction of $LTP_{E\rightarrow I}$ was blocked by FAC but not in control slice **(Fig. 1 D-F)**. Next, we tested whether glial metabolism is involved in $LTP_{E\rightarrow I}$ in CA1 stratum orient, which has been shown to depend on calcium-permeable AMPARs (CP-AMPARs) and metabotropic glutamate receptors (mGluRs) (13). We found that the induction of $LTP_{E\rightarrow I}$ is not blocked by FAC in CA1 stratum oriens (**Fig. S4 A-B**). We repatched 6 of these cells (4 cells from control group and 2 cells from FAC treated group) in whole-cell voltage-clamp mode and observed that they exhibited high rectification AMPARs and a small component of NMDAR-mediated current (**Fig. S5**). This observation confirms previous study findings (32, 34). Moreover, we found that the induction of $LTP_{E\rightarrow I}$ is not blocked by D-AP5 in CA1 stratum oriens (**Fig. S4**). Overall, the results indicate that the formation of $LTP_{E\rightarrow I}$ in CA1 stratum radiatum is tightly regulated by astrocyte function. The results in stratum oriens also exclude the possibility that FAC directly affects the metabolism of interneuron which inhibit the formation of $LTP_{E\rightarrow I}$.

It is commonly accepted that astrocytic calcium signaling plays a pivotal role in triggering the release of gliotransmitters and modulating synaptic transmission (24, 35–38). In addition, it is well documented that astrocytic calcium signaling is necessary to the formation of $LTP_{E\rightarrow E}$ in CA1 stratum radiatum (4, 26). Thus, we tested whether intracellular Ca^{2+} signals are required for the induction of $LTP_{E\rightarrow I}$ in CA1 stratum radiatum. We found that clamping of astrocyte Ca^{2+} concentration significantly suppressed $LTP_{E\rightarrow I}$ **(Fig. 2 A-D)**. Consistent with the previous study in $LTP_{E\rightarrow E}$, supply of D-serine (50 μM) fully rescued NMDAR-dependent $LTP_{E\rightarrow I}$ **(Fig. 2D)**. For the control, when the intracellular Ca^{2+} concentration of astrocytes was not clamped, but the astrocyte was recorded with a glass pipette, $LTP_{E\rightarrow I}$ was indistinguishable from that induced without patching an astrocyte **(Fig. 2D)**. Overall, these results indicate that functional preservation of astrocytic metabolism and Ca^{2+} mobilization is critical for maintaining the induction of $LTP_{E\rightarrow I}$.

Astrocytic Ca^{2+} transients, induced by activation of astroglial cannabinoid type 1 receptors (CB1Rs), are involved in the regulation of $LTP_{E\rightarrow I}$ formation

Accumulating evidence indicates that neuronal depolarization in the hippocampus induced astrocytic Ca^{2+} transients, which is mediated by the activation of astroglial CB1Rs (39–43). Moreover, it has been shown that astroglial CB1R-mediated Ca^{2+} elevation is necessary for $LTP_{E\rightarrow E}$ in the CA1 region of hippocampus (26). Therefore, we asked whether astroglial CB1Rs-mediated Ca^{2+} elevations are required for hippocampal $LTP_{E\rightarrow I}$. We first analyzed whether TBS could evoke astrocytic Ca^{2+} transient via CB1Rs in the stratum radiatum of hippocampus. In this study, GCaMP6f was used to analyze astrocytic Ca^{2+} signal, which were specifically expressed in astrocytes by using adeno-associated viruses of the 2/5 serotype (AAV 2/5) with the astrocyte-specific gfaABC$_1$D promoter **(Fig. S6 A)**. Furthermore, the expression was confirmed by immunohistochemistry. Within the virally transduced region, GCaMP6f expression positive cells in CA1 stratum radiatum were also positive to the astrocyte specific marker GFAP **(Fig. S6 A, C-D)**. Co-staining with the neuron marker NeuN showed no overlap with GCaMP6f expression **(Fig. S6 B, E)**. Consistent with previously studies (26, 44), we found that TBS significantly increased the Ca^{2+} signaling in astrocytes of hippocampal slices in the presence of picrotoxin and CGP55845 **(Fig. 3 A-B, F)**. As predicted, the increase in Ca^{2+} signals after TBS were inhibited by the CB1 receptor inhibitor AM251 **(Fig. 3 C-D, F)**. Previous studies have demonstrated that activation of

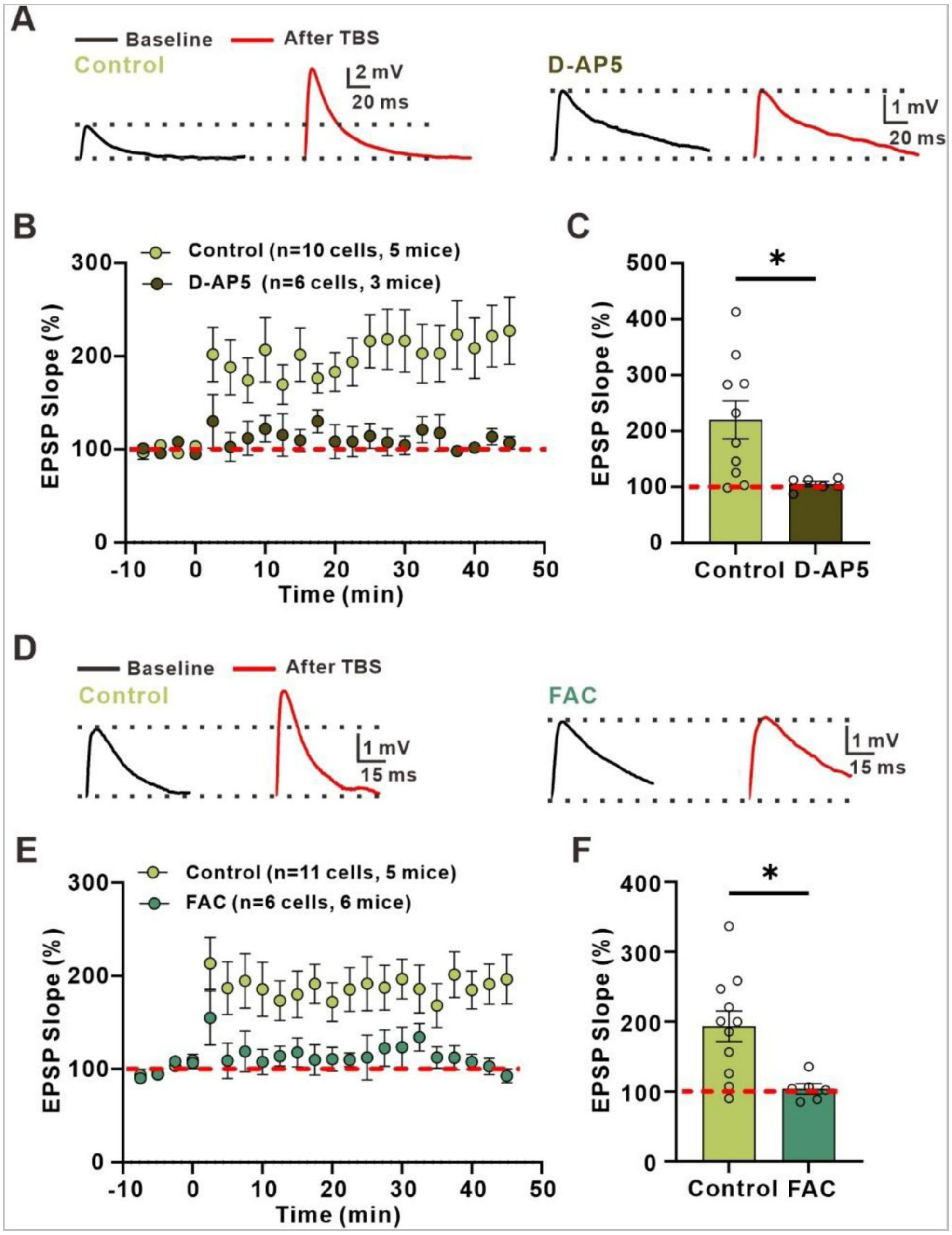

Fig. 1.

LTP$_{E \rightarrow I}$ in stratum radiatum is dependent on the activation of NMDA receptor and astrocytic metabolism.

(A) Left: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) LTP induction; Right: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) LTP induction while in the presence of NMDA receptor antagonist D-AP5.
(B) Normalized slope before and after the TBS stimulation protocol in control conditions and in the presence of NMDA receptor antagonist D-AP5.
(C) The summary data measured 35-45 min after LTP induction **(Control: 220.2±33.94, n=10 slices from 5 mice; D-AP5: 105.2±4.404, n=6 slices from 3 mice; t=2.580 with 14 degrees of freedom, *p*=0.0218, two-tailed unpaired t-test**).
(D) Left: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) LTP induction; Right: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) LTP induction while astrocytic metabolism was disrupted.
(E) Normalized slope before and after the TBS stimulation protocol in control conditions and in the presence of astrocytic metabolism inhibitor FAC.
(F) The summary data measured 35-45 min after LTP induction **(Control: 193.5±21.84, n=11 slices from 5 mice; FAC: 103.8±7.278, n=6 slices from 6 mice; t=2.943 with 15 degrees of freedom, *p*=0.0101, two-tailed unpaired t-test).**

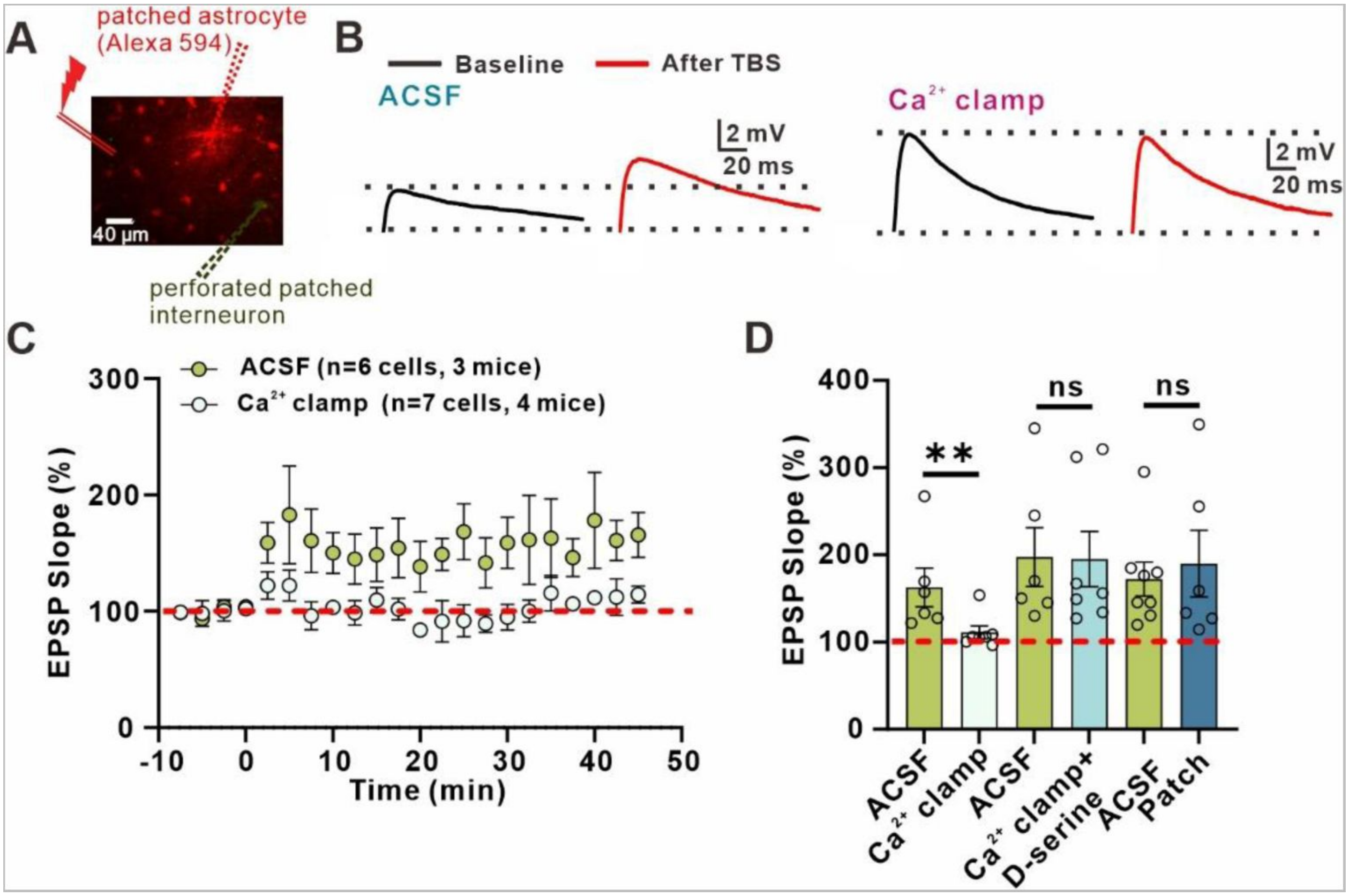

Fig. 2.

Astrocyte Ca^{2+} is involved in the induction of $LTP_{E\rightarrow I}$.

(A) Representative image of the astrocytic syncytium illustrating that Ca^{2+} clamping reagents and Alexa Fluor 594 diffuse through gap junctions surround the patched interneuron.
(B) Left: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) LTP induction; Right: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) LTP induction when Ca^{2+} concentration was clamped.
(C) Normalized slope before and after the TBS stimulation protocol in control conditions and when Ca^{2+} concentration was clamped.
(D) The summary data measured 35-45 min after LTP induction (**ACSF: 162.7±22.21, n=6 slices from 3 mice; Ca^{2+} clamp: 111.1±7.323, n=7 slices from 4 mice; Mann-Whitney U Statistic= 3.000, *p*=0.008, Mann-Whitney Rank Sum Test; ACSF: 197.5±33.83, n=6 slices from 4 mice; Ca^{2+} clamp+D-serine: 195.1±31.75, n=7 slices from 4 mice; Mann-Whitney U Statistic= 19.000, *p*=0.836, Mann-Whitney Rank Sum Test; ACSF: 172.2±19.49, n=8 slices from 4 mice; patch: 189.9±38.11, n=6 slices from 3 mice; Mann-Whitney U Statistic= 22.000, *p*=0.852, Mann-Whitney Rank Sum Test**).

α1-adrenoceptors increases calcium signals (1, 45–49) and triggers the release of D-serine release in the neocortex (50). However, our results show that α1-adrenoceptors receptors are not involved in the Ca^{2+} signal increase observed after TBS **(Fig. 3F)**.

Next, we explore whether CB1Rs-mediated Ca^{2+} elevation was accompanied by the formation of $LTP_{E\rightarrow I}$. We found that $LTP_{E\rightarrow I}$ was significantly reduced in AM251 treated slice relative to controls **(Fig. 3 G-I)**. Consistent with the above results shown in **Fig. 2D**, $LTP_{E\rightarrow I}$ was rescued by the addition of D-serine in AM251-treated slices

D-serine release from astrocyte potentiates NMDAR-mediated synaptic response

Next, we explored the underlying mechanisms by which astrocytes control the formation of $LTP_{E\rightarrow I}$ in the CA1 stratum radiatum. Our above results indicate that D-serine is a downstream signal pathway of astrocyte Ca^{2+} signal. D-serine, a co-agonist of NMDAR, could be released by astrocytes through Ca^{2+}-dependent exocytosis and regulate the function of NMDAR. It has been shown that the occupancy of synaptic NMDAR co-agonist sites by D-serine is not saturated in CA1 pyramidal cells (26, 51). However, the level of occupancy of synaptic NMDAR co-agonist sites by D-serine in interneurons of CA1 stratum radiatum remains unclear. Furthermore, previous studies have demonstrated that NMDAR-dependent synaptic responses of interneurons in the stratum radiatum display a strong rundown effect under whole-cell mode (31). Thus, we recorded the NMDAR-mediated EPSPs from stratum radiatum interneurons in the perforated-patch configuration in Mg^{2+} free ACSF, which showed no rundown effect in this configuration **(Fig. 4 A-B)**. Bath application of 50 μM D-serine enhanced the NMDAR-mediated EPSPs **(Fig. 4 A-C)**, indicating that the level of D-serine in the excitatory to inhibitory synapse cleft was not saturated to occupy the co-agonist site. Subsequently, we tested whether TBS could increase extracellular level of D-serine and promote NMDAR-mediated synaptic responses. Our results indicated that NDMAR-mediated responses were transiently enhanced after one TBS, and the percentage of potentiation was reduced by pre-treatment hippocampal slice with 50 μM D-serine **(Fig. 4 D-F)**. However, disrupting glial metabolism with FAC rendered the percentage of TBS-induced potentiation insensitive to bath application of D-serine **(Fig. 4F)**. Furthermore, the percentage of potentiation in AMPAR-mediated responses after TBS is insensitive to bath application of D-serine, indicating that the enhancement of NMDAR-mediated response by TBS is not due the change of release probability and cell excitability **(Fig. 4F)**. Taken together, our results indicate that D-serine release from astrocyte potentiates NMDAR-mediated responses by binding the co-agonist site and regulates the formation of $LTP_{E\rightarrow I}$.

Chemogenetic activation of astrocytes induced E→I synaptic potentiation

The crucial role of astrocytes in $LTP_{E\rightarrow I}$ has been extensively demonstrated in brain slices (Henneberger et al., 2010; Min and Nevian, 2012; Pascual et al., 2005; Perea and Araque, 2007; Suzuki et al., 2011) and in vivo (26). In this study, we aimed to investigate if the activation of astrocytic G protein-coupled receptor (GPCR) pathway could trigger potentiation of E→I synaptic transmission. We utilized an adeno-associated virus serotype 5 (AAV2/5) vector encoding hM3Dq fused to mCherry for the specific activation of astrocytes via clozapine-N-oxide (CNO). To ensure specific expression in astrocytes, the vector was also under the control of the astrocyte-specific gfaABC1D promoter **(Fig. S7)**.

To verify whether CNO application could evoke Ca^{2+} transients in astrocytes, we delivered AVV of hM3Dq along with AAV of GCaMP6f, and conducted confocal Ca^{2+} imaging in brain slices. Our results showed that CNO application indeed induced an increase in intracellular Ca^{2+} levels in

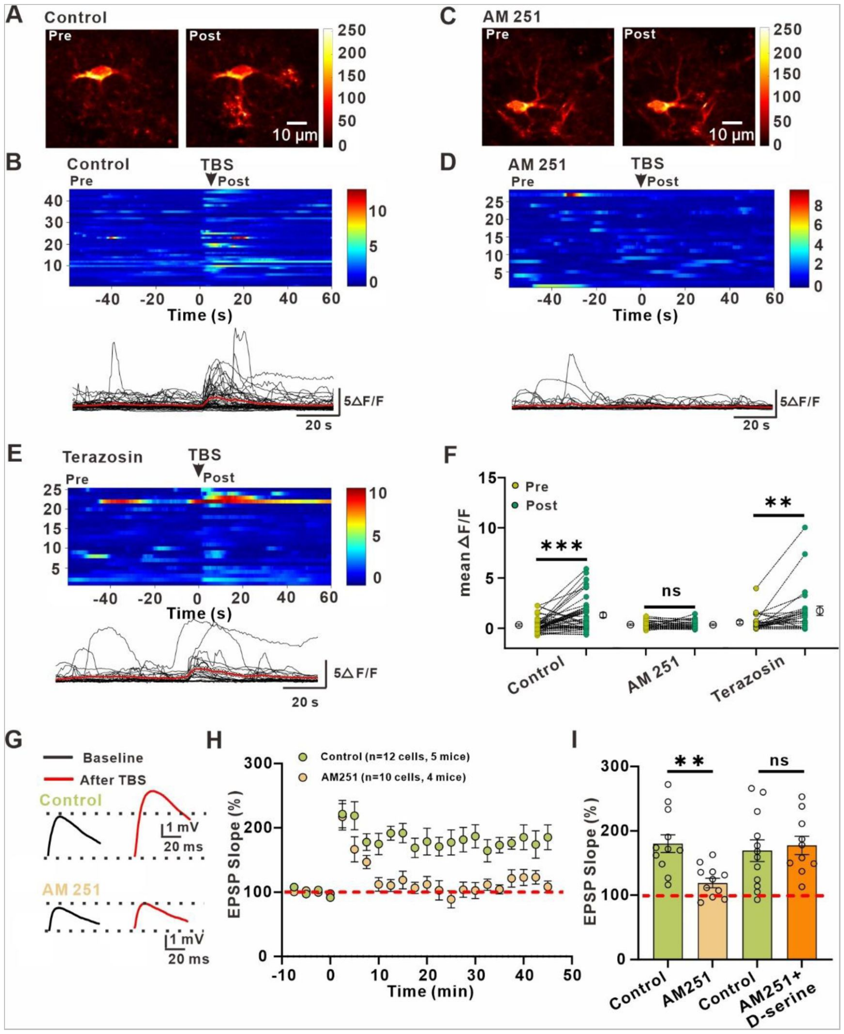

Fig. 3.

Activation of astrocytic CB1 receptors causes an increase in astrocytic Ca^{2+} signals and is involved in the induction of $LTP_{E \to I}$.

(A) Representative images of a GCaMP6f^{+} astrocyte before (left) and after (right) theta burst stimulation (TBS).
(B) Kymographs and △F/F traces of cells with Ca^{2+} signals evoked by the activation of Schaffer collateral with a TBS in GCaMP6f^{+} astrocyte.
(C) Representative images of a GCaMP6f^{+} astrocyte before (left) and after (right) theta burst stimulation (TBS) when in the presence of CB1 receptor blocker AM251. .
(D) Kymographs and △F/F traces of cells with Ca^{2+} signals evoked by the activation of Schaffer collateral with a TBS in GCaMP6f^{+} astrocyte when in the presence of CB1 receptor blocker AM251.
(E) Kymographs and △F/F traces of cells with Ca^{2+} signals evoked by the activation of Schaffer collateral with a TBS in GCaMP6f^{+} astrocyte when in the presence of α1-adrenoceptor blocker terazosin.
(F) Summary plots for experiments in (B)-(E) illustrating that Ca^{2+} signals elicited by TBS is mediated by the activation of CB1 receptors **(Control pre: 0.3275±0.1083 △ F/F, Control post: 1.313±0.2483 △F/F, n=45 from 20 cells of 5 mice; z= 3.369, *p*<0.001, Wilcoxon Signed Rank Test; AM251 pre: 0.3541±0.07219 △F/F, AM251 post: 0.3418±0.07240 △F/F, n=28 from 13 cells from 5 mice; z= −0,797, *p*=0.432, Wilcoxon Signed Rank Test; Terazosin pre: 0.5725±0.1749 △F/F, Terazosin post: 1.73±0.4661 △F/F, n=25 from 11 cells from 4 mice; z=3.215, *p*=0.001, Wilcoxon Signed Rank Test**).
(G) Upon: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) LTP induction; below: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) LTP induction while in the presence of CB1 receptor antagonist AM251.
(H) Normalized slope before and after the TBS stimulation protocol in control conditions and in the presence of CB1 receptor antagonist AM251.
(I) The summary data measured 35-45 min after LTP induction (**Control: 180.2±13.76, n=12 slices from 5 mice; AM251: 118.9±7.406, n=10 slices from 4 mice; t=3.697 with 20 degrees of freedom, *p*=0.0014, two-tailed unpaired t-test; Control: 169.3±16.89, n=12 slices from 6 mice; AM251+D-serine: 177.4±14.27, n=10 slices from 5 mice; t=0.3561 with 20 degrees of freedom, *p*=0.7255, two-tailed unpaired t-test**).

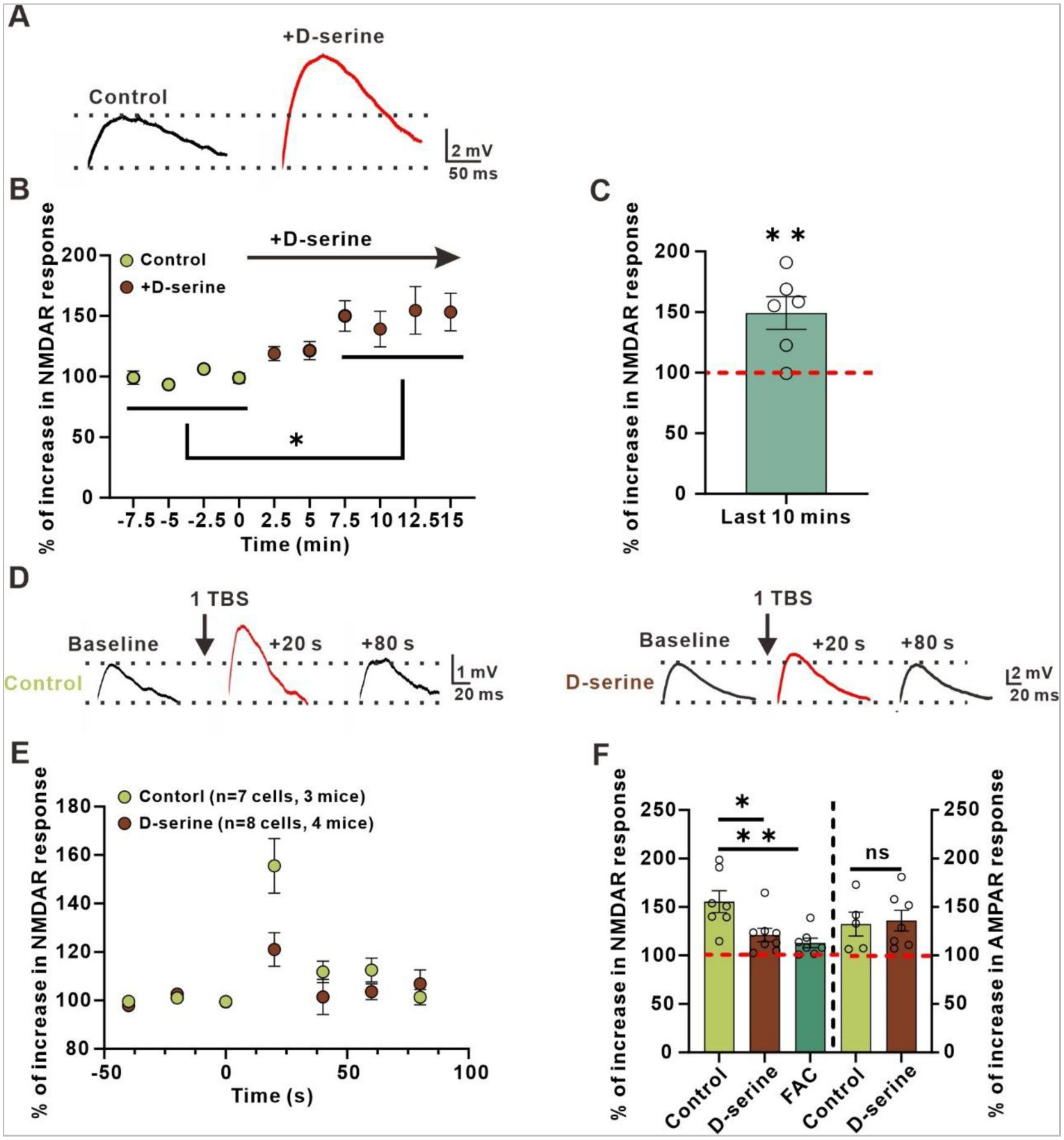

Fig. 4.

D-serine release from astrocyte potentiates NMDAR-mediated responses

(A) Example traces of NMDAR-mediated EPSPs before and after bath application of D-serine in stratum radiatum interneurons.

(B) Normalized amplitude before and after add D-serine in stratum radiatum interneurons

(C) The percentage of potentiation in NMDAR-mediated responses **(percentage of potention: 155.523 ± 11.234, n=6 slices from 3 mice; t=-4.942 with 6 degrees of freedom, p=0.00260, two-tailed paired t-test**).

(D) Left: example traces of NMDAR-mediated EPSPs before and after the TBS stimulation protocol in stratum radiatum interneurons .Right: example traces of NMDAR-mediated EPSPs before and after the TBS stimulation protocol in stratum radiatum interneurons when in the presence of D-serine.

(E) Normalized amplitude before and after the TBS stimulation protocol in stratum radiatum interneurons during ASCF and after application D-serine.

(F) The summary data measured the percentage of potentiation in NMDAR-mediated responses (**Control: 155.5±11.23, n=7 slices from 3 mice; D-serine: 121.1±6.917, n=8 slices from 4 mice; FAC: 112.8±4.931, n=7 slices from 3 mice; *p*=0.0036, F(2, 19)=7.653, ANOVA with Dunnett's comparison) and AMPAR-mediated responses after TBS(Control: 132.357±12.305, n=5 slices from 3 mice; D-serine: 136.012±10.611, n=8 slices from 4 mice; t=-0.224 with 10 degrees of freedom, p=0.827, two-tailed unpaired t-test**).

cells co-expressing hM3Dq and GCaMP6f **(Fig. 5 A-D)**. These results suggest that the expression of hM3Dq is selective to astrocytes and can elicit a rise in intracellular Ca^{2+} levels upon administration of CNO.

Subsequently, we investigated the impact of astrocytic Gq activation on evoked synaptic events in interneurons of CA1 stratum radiatum induced by Schaffer Collaterals stimulation, both before and after the administration of CNO. Interestingly, we observe that the EPSC amplitude was potentiated by 60% in response to the exact same stimulus in gfaABC1D::hM3Dq slices treated with CNO **(Fig. 5 E-G)**, while no such potentiation was detected in slices obtained from the mice that were injected a control virus (AAV2/5-gfaABC1D::mCherry) **(Fig. 5 E-G)**.

Previous studies have indicated that the synaptic potentiation triggered by chemogenetic activation of astrocyte is mediated through the release of D-serine by astrocytes, resulting in the activation of NMDARs (52). To verify whether the astrocytic-induced synaptic potentiation between excitatory and inhibitory neurons is mediated by D-serine release from astrocytes and the subsequent activation of NMDARs, we conducted an experiment in which we administered CNO after blocking the NMDARs with D-AP5 or saturating the glycine site of NMDARs with 50 μM D- serine. Our results showed that both the NMDAR blocker D-AP5 and 50 μM D-serine completely inhibited the potentiation in EPSP amplitude observed in response to CNO- induced astrocytic activation **(Fig. 5 H-J)**. Our findings demonstrate, for the first time, that astrocytic activation alone can trigger *de novo* potentiation of synapses between excitatory and inhibitory neurons, and that this potentiation is indeed mediated by the release of D-serine from astrocytes and subsequent activation of NMDARs.

$LTP_{E \to I}$ in CA1 of stratum radiatum is necessary for long-term memory formation

We have previously shown that GABAergic interneurons in hippocampus express high levels of γCaMKII, while αCaMKII, βCaMKII and δCaMKII are expressed at a lower frequency (53). Additionally, studies have demonstrated that γCaMKII expressed in hippocampal parvalbumin positive (PV^+) interneurons and cultured hippocampal inhibitory interneurons is essential for the induction of $LTP_{E \to I}$ (53, 54). Specifically, in hippocampal PV^+ interneurons, this protein is also vital for the formation of hippocampal-dependent long-term memory in vivo (53). Therefore, we asked whether astrocyte gated $LTP_{E \to I}$ in CA1 stratum radiatum involved in the hippocampal- dependent long-term memory formation. Several independent groups provide compelling evidence that astroglial CB1R-mediated signaling pathway regulates excitatory synapse formed by excitatory neurons, rather than excitatory synapses between excitatory neurons and interneurons (39–42, 55–57). Thus, to manipulate this pathway may also affect excitatory synapse formed by excitatory neurons. To avoid this side effect, we specifically knockdown of γCaMKII expression in inhibitory neurons of CA1 stratum radiatum by delivering an adeno-associated virus serotype 2/9 (AAV2/9) vector encoding shRNAs against γCaMKII and EGFP under the control of an inhibitory neuron-specific promoter (AAV2/9-mDLx-EGFP-γCaMKII shRNA) through bilateral stereotactic injection. We observed that interneurons identified by the specific markers GAD67 were also positive for EYFP and therefore likely expressed shRNAs, which led to knockdown γCaMKII in these cells **(Fig. S8)**. To confirm that knockdown γCaMKII could hamper the induction of $LTP_{E \to I}$, the EYFP positive interneurons in CA1 stratum radiatum were recorded in perforated-patch mode. We found that knocking down γCaMKII in CA1 stratum radiatum interneurons impaired $LTP_{E \to I}$, but $LTP_{E \to I}$ in the putative interneurons in stratum radiatum of WT mice is unimpaired **(Fig. 6 A-C)**. In addition, robust $LTP_{E \to I}$ induced by TBS in EYFP positive interneurons injected with AAV-mDLx-scramble shRNA into the CA1 stratum radiatum of hippocampus **(Fig. 6C)**. Moreover, it is worth noting that the resting membrane potential, frequency and amplitude of sEPSC, excitability and PPR recorded from γCaMKII knockdown interneurons did not differ from those recorded from putative interneurons and EYFP

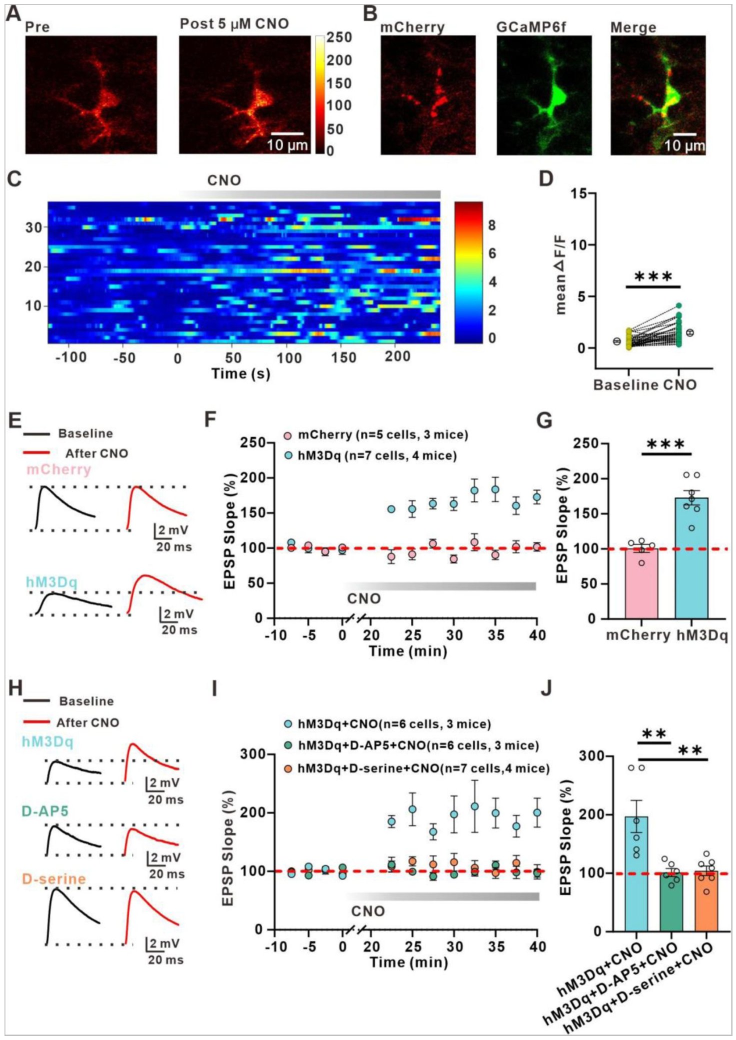

Fig. 5.

Astrocytic activation induces *De Novo* LTP$_{E\rightarrow I}$.

(A) Representative images of a GCaMP6f$^+$ astrocyte before (left) and after (right) CNO application.

(B) Example confocal images which showing co-expression of GCaMP6f and mCherry constructs confirm that representative cell in (A) is co-expression hM3Dq.

(C) Kymographs and ΔF/F traces of cells with Ca^{2+} signals evoked by bath application of CNO in GCaMP6f$^+$ and mCherry$^+$ astrocyte.

(D) Summary plot illustrating that CNO is effective to activate astrocytes **(Baseline: 0.6167±0.07762** ΔF/F, **CNO: 1.456±0.1504** ΔF/F, **n=36 from 19 cells of 3 mice; t=6.411 with 35 degrees of freedom, *p*<0.0001, two-tailed paired t-test)**.

(E) Upon: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) CNO application when astrocyte only express mCherry; below: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) CNO applicaion when hM3Dq expressed astrocytes were activated by CNO

(F) Relative EPSP slope before and after CNO application when astrocytes only express mCherry and when astrocytes express hM3Dq.

(G) The summary data measured 35-45 min after LTP induction (**mCherry: 100.6±5.651, n=5 slices from 3 mice; hM3Dq: 172.8±10.28, n=7 slices from 4 mice; t=5.474 with 10 degrees of freedom, *p*=0.0003, two-tailed unpaired t-test**).

(H) Upon: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-40 min after (red traces) CNO application; middle: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-40 min after (red traces) CNO application when in the presence NMDA receptor blocker D-AP5; below: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-40 min after (red traces) CNO application when the glycine sites of NMDA receptors were saturated by D-serine.

(I) Relative EPSP slope before and after CNO application when astrocytes were activated by CNO, and when in the presence of NMDA receptor antagonist D-AP5 and in the presence of D-serine.

(J) The summary data measured 35-45 min after LTP induction **(hM3Dq+CNO: 197.1±27.57, n=6 slices from 3 mice; hM3Dq+CNO+D-AP5: 101.3±6.886, n=6 slices from 3 mice; hM3Dq+CNO+D-serine: 104.2+8.166, n=7 slices from 4 mice; *p*=0.0011, F(2, 16)=10.80, ANOVA with Dunnett's comparison).**

positive interneurons (infected with scramble shRNA) **(Fig. S9)**. Taken together, these results indicate that knocking down γCaMKII from interneuron has no effect on their synaptic transmission at baseline, but impaired the induction of $LTP_{E \to I}$ in these interneurons.

Next, we examined the behavioral consequences of destroying astrocyte gated $LTP_{E \to I}$ in CA1 stratum radiatum. We analyzed contextual fear conditioning memory, which is associated with activation of the hippocampus. We found that no effect of γCaMKII knockdown on exploration of the context before conditioning was observed **(Fig. 6 D-E)**. 24 hours after training, mice were returned to the training box, and freezing was measured at the first 2 min. We found that γCaMKII knockdown mice showed a significantly reduction in freezing during contextual conditioning **(Fig. 6F)**. Consistent with our previous study (53), γCaMKII knockdown in interneurons in CA1 stratum radiatum did not produce a significant effect on freezing during tone conditioning at 28 h after training **(Fig. 6G)**. Taken together, our results strongly suggest that astroglial CB1R signaling pathway gated $LTP_{E \to I}$ in CA1 stratum radiatum plays a vital role in hippocampus-dependent long-term memory.

Discussion

In the present study, we identify $LTP_{E \to I}$ in CA1 stratum radium is tightly control by D-serine release from astrocyte via the astroglial CB1Rs-mediated Ca^{2+} elevation. In addition, knockdown of γCaMKII by specific shRNA in interneurons in CA1 stratum radium causes cognitive function deficit. Taken together, out data support that astrocyte gated $LTP_{E \to I}$ in CA1 stratum radium plays a critical role in preserve normal cognitive function.

CB1Rs are widely expressed in various brain regions including hippocampus and detectable in presynaptic terminals (56, 58), postsynaptic terminals (59, 60), intracellular organelles (61, 62) and also on astrocytes (26, 39–42, 55, 57, 63). However, the effects of astroglial CB1R-mediated signaling on synaptic transmission and plasticity is highly debated. It has been shown that exogenous administration of Δ^9- tetrahydro-cannabinol (THC) lead to a temporally prolonged and spatially widespread activation of astroglial CB1 receptors and trigger glutamate release, which activates postsynaptic NMDARs and induces LTD in the CA3–CA1 hippocampal synapses, resulting working memory deficit (57). Araque and colleagues reported that eCB released from depolarized CA1 pyramidal cell activates astroglial CB1Rs with a shorter and localized way and induces glutamate release, which activates lateral presynaptic mGluRs and induces LTP (41, 42). In another study, Robin et al found that high frequency stimulation (HFS) of Schaffer Collateral induces LTP, which is gated by the activation of astroglial CB1Rs and the release of D-serine from astrocytes (26). These diverse consequences by activation of CB1 receptor may be due to different neuronal activity pattern that induce eCB release and different nature of the agonists. Interestingly, it has been discovered that individual hippocampal astrocytes are capable of releasing both ATP/adenosine and glutamate, and that this release occurs in a time-dependent and activity-sensitive manner in response to neuronal interneuron activity (64). These findings suggest that the specific type and intensity of astrocyte stimulation plays a critical role in determining the downstream signaling pathways that are triggered by CB1R activation in astrocytes. Consistent with previous study, we found that the activation of astroglial CB1Rs induces Ca^{2+} elevation and triggers the release of D-serine that binds to postsynaptic NMDAR and induces LTP formation (26). Our results provide evidence that astroglial CB1R-mediated singling not only modulates the E→E synapses, but also regulates E→I synapses.

The precise mechanisms through which neurons and astrocytes differentially regulate D-serine levels are yet to be fully elucidated (65–67). However, it is evident that astrocytes play a significant role in regulating the availability of D-serine. The enzyme serine racemase catalyzes the conversion of L-serine into D-serine, which was initially found in astrocytes and microglia in the mammalian brain (29, 68, 69). It should be highlighted that serine racemase has also

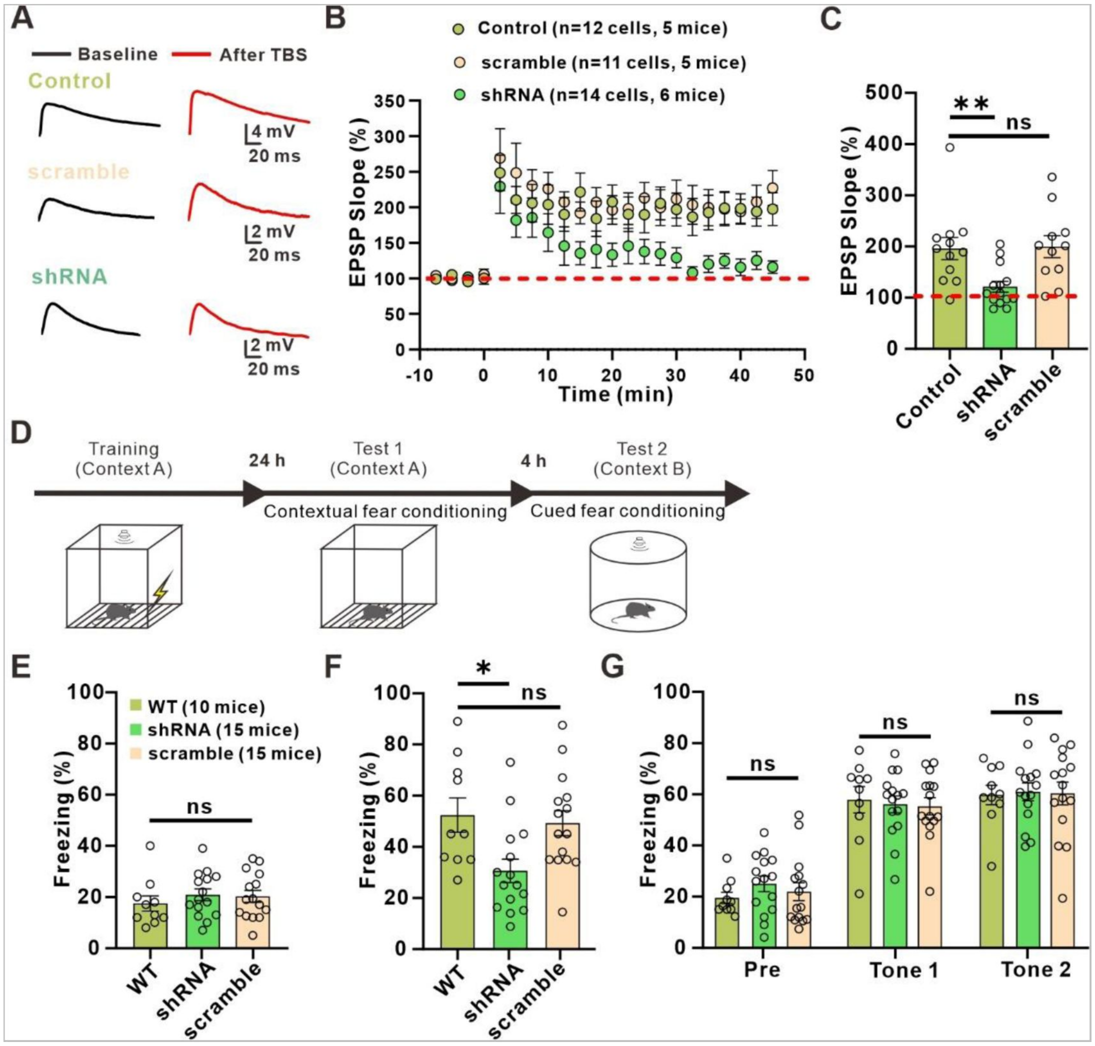

Fig. 6.

Impaired hippocampus-dependent long-term memory in knockdown γCaMKII mice

(A) Upon: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) LTP induction in WT group; middle: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) LTP induction in AAV-mDLx-scramble shRNA group; below: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) LTP induction in AAV-mDLx-γCaMKII shRNA group.
(B) Normalized slope before and after the TBS stimulation protocol in control, scramble shRNA and γCaMKII shRNA group.
(C) The summary data measured 35-45 min after LTP induction **(Control: 196±21.56, n=12 slices from 5 mice; γCaMKII shRNA: 121.3±10.64, n=14 slices from 6 mice; scramble shRNA: 199.6±21.42, n=11 slices from 5 mice; *p*=0.0040, F(2, 34)=6.527, ANOVA with Dunnett's comparison).**
(D) Schematic illustration of the experimental design for contextual and cued fear conditioning test, indicating the time line of the experimental manipulations.
(E) The freezing responses was measured in context A before training in WT, bilaterally injected γCaMKII shRNA and scramble shRNA mice **(Control: 17.5±2.975, n=10 mice; γCaMKII shRNA: 20.9±2.275, n=15 mice; scramble shRNA: 20.29±2.341, n=15 mice; *p*=0.6378, F(2, 37)=0.4553, ANOVA with Dunnett's comparison).**
(F) The freezing responses was measured 24 h after training in WT, bilaterally injected γCaMKII shRNA and scramble shRNA mice **(Control: 52.4±6.695, n=10 mice; γCaMKII shRNA: 30.64±4.574, n=15 mice; scramble shRNA: 49.27±4.899, n=15 mice; *p*=0.0105, F(2, 37)=5.170, ANOVA with Dunnett's comparison).**
(G) The freezing response was measured 28 h after training in WT, bilaterally injected γCaMKII shRNA and scramble shRNA mice **(Pre Control: 19.5±2.171, n=10 mice; Pre γCaMKII shRNA: 25.01±3.016, n=15 mice; Pre scramble shRNA: 21.94±3.519, n=15 mice; *p*=0.4986, F(2, 37)=0.7093, ANOVA with Dunnett's comparison; Tone 1 Control: 57.9±5.153, n=10 mice; Tone 1 γCaMKII shRNA: 56.09±3.319, n=15 mice; Tone 1 scramble shRNA: 55.32±3.368, n=15 mice; *p*=0.9002, F(2, 37)=0.1055, ANOVA with Dunnett's comparison; Tone 2 Control: 59.75±3.825, n=10 mice; Tone 2 γCaMKII shRNA: 60.95±3.482, n=15 mice; Tone 2 scramble shRNA: 60.34±4.485, n=15 mice; *p*=0.9802, F(2, 37)=0.01996, ANOVA with Dunnett's comparison).**

been detected in neurons (70–72). A study showed that, despite a significant reduction in SR protein levels in the brains of neuronal SR knockout mouse brains, the reduction in D-serine levels was minimal, suggesting that neurons are not the exclusive source of D-serine (70) and neurons may produce and release D-serine under certain conditions. Notably, activation of G protein-coupled receptors in astrocytes through chemogenetic means results in increased synaptic transmission, which is dependent on the availability of D-serine (52, 73). Specifically, the release of D-serine, which is regulated by CB1R in astrocytes, is required for Ca^{2+}-dependent modulation of LTP in vivo (26), as well as the threshold and amplitude of dendritic spikes (74). Moreover, recent studies have shown that conditional connexin double knockout (75) or knockdown of α4nAChR (76) in astrocytes can decrease the extracellular concentration of D-serine, which in turn reduces NMDAR-dependent synaptic potentiation. These findings suggest that astrocytes are the primary source of D-serine, which plays a crucial role in modulating the function of NMDARs.

It is well established that $LTP_{E\rightarrow E}$ observed in the CA1 region of the hippocampus is triggered by activation of NMDARs. However, $LTP_{E\rightarrow I}$ is less studied compared to $LTP_{E\rightarrow E}$, but recent evidence suggests that it also involves the activation of NMDARs (10, 11, 31, 32, 53, 77). There have been reports of NMDAR-dependent $LTP_{E\rightarrow I}$ in various regions of the brain, including the hippocampus and cortex (10, 11). Our results suggest that different synaptic mechanisms are involved in the induction of $LTP_{E\rightarrow I}$ in different subregions of the hippocampus. The stratum radiatum, where interneuorns contain NMDARs, is known to be sensitive to NMDAR-dependent $LTP_{E\rightarrow I}$. In contrast, the stratum oriens, where interneuorns contain CP-AMPARs, appears to rely on the activation of CP-AMPA receptors for $LTP_{E\rightarrow I}$ induction. Notably, our findings suggest that astrocytes contribute to NMDAR signaling in the induction of $LTP_{E\rightarrow I}$ in the stratum radiatum through the modulation of extracellular levels of the co-agonist D-serine. It is worth noting that our study found that prolonged activation of astrocytes via the Gq-DREADD pathway resulted in a substantial and persistent increase in Ca^{2+} events and significantly potentiated EPSP responses. This is in consistent with earlier observations made by other groups regarding $LTP_{E\rightarrow E}$ (52, 73). Above all, the mechanism of $LTP_{E\rightarrow I}$ in stratum radiatum appears to be share by $LTP_{E\rightarrow E}$ observed in the CA1 region.

A previous study has demonstrated that knocking down γCaMKII from interneurons can disrupt $LTP_{E\rightarrow I}$ and cognitive function (53, 54). Our results confirmed that knocking down γCaMKII in interneurons of the stratum radiatum also leads to disruption of $LTP_{E\rightarrow I}$ and cognitive function. Ma and colleagues showed that hippocampal network oscillations in the gamma and theta bands were significantly weaker in γCaMKII knockout mice compared to wild-type mice, following learning (53). This suggests that impaired experience-dependent oscillations in the hippocampus of γCaMKII PV-KO mice may lead to cognitive dysfunction. In this respect, it will be intriguing to investigate the network oscillation after learning in our condition in the future study.

In the hippocampus, GABAergic local circuit inhibitory interneurons make up approximately 10-15% of the total neuronal cell population (78). However, these interneurons are diverse in their subtypes, morphology, distribution, and functions (15, 79). In our study, we mainly focused on a subpopulation of interneurons in the stratum radiatum of hippocampus. Although it is unclear which type of interneuron recorded in our study, but our study indicated that most of interneurons in the stratum radiatum do not express CP-AMPARs but express abundant of NMDARs. These findings are in line with a previous study conducted by Lasmsa *et al.* (32). In our study, we observed that approximately 80% of interneurons in stratum radium were able to induce LTP successfully. This finding contrasts with the observation made by Lamsa *et al.*, who reported a figure of approximately 52% interneurons capable of inducing LTP. The reason for this discrepancy could be attributed to differences in the induction protocol used in the respective studies.

Our study corroborates earlier research that suggests distinct synaptic mechanisms are involved in LTP induction in the CA1 region of the hippocampus across different subregions (11, 13, 32). However, the major breakthrough of our study is the demonstration that astrocytic function serves as the gating mechanism for LTP $_{E\rightarrow I}$ induction in the stratum radiatum. Additionally, our data reveals that the activation of astrocytes via the Gq-DREADD pathway produces *de novo* long-lasting potentiation of EPSP in stratum radiatum interneurons and the knockdown of γCaMKII disrupts cognitive function. These results shed light on the complex mechanisms underlying learning and memory in the hippocampus, and may have implications for developing new therapies targeted at modulating astrocytic function for the treatment of memory disorders.

Acknowledgements

This work was supported by grants from key R&D Program Project of Zhejiang (2022C03034), Zhejiang Provincial Natural Science Foundation of China (LQ23C090001), Scientific Research Foundation of Zhejiang University City College, (X-202103), Scientific Research Foundation of Hangzhou City University (J-202325), Science and Technology Department of Zhejiang Province (2021RC051).

Conflict of interest

The authors declare no competing interests.

Data availability statement

Data are available from the corresponding author on reasonable request.

Author contributions

W.S. and L.Z. designed the study; W.S. Y.T. and L.Z. analyzed data, wrote the manuscript; W.S., Y.T. J.Y. and L.Z. carried out most experiments; F.Z. and W.Z. helped with viral injection; Y.X. and J.D. helped with immunohistochemistry; J.D. and F.Z. helped with slice electrophysiology; W.Z. and L.X. helped with data analysis. All authors discussed the results and comments on the manuscript.

Supporting Figures

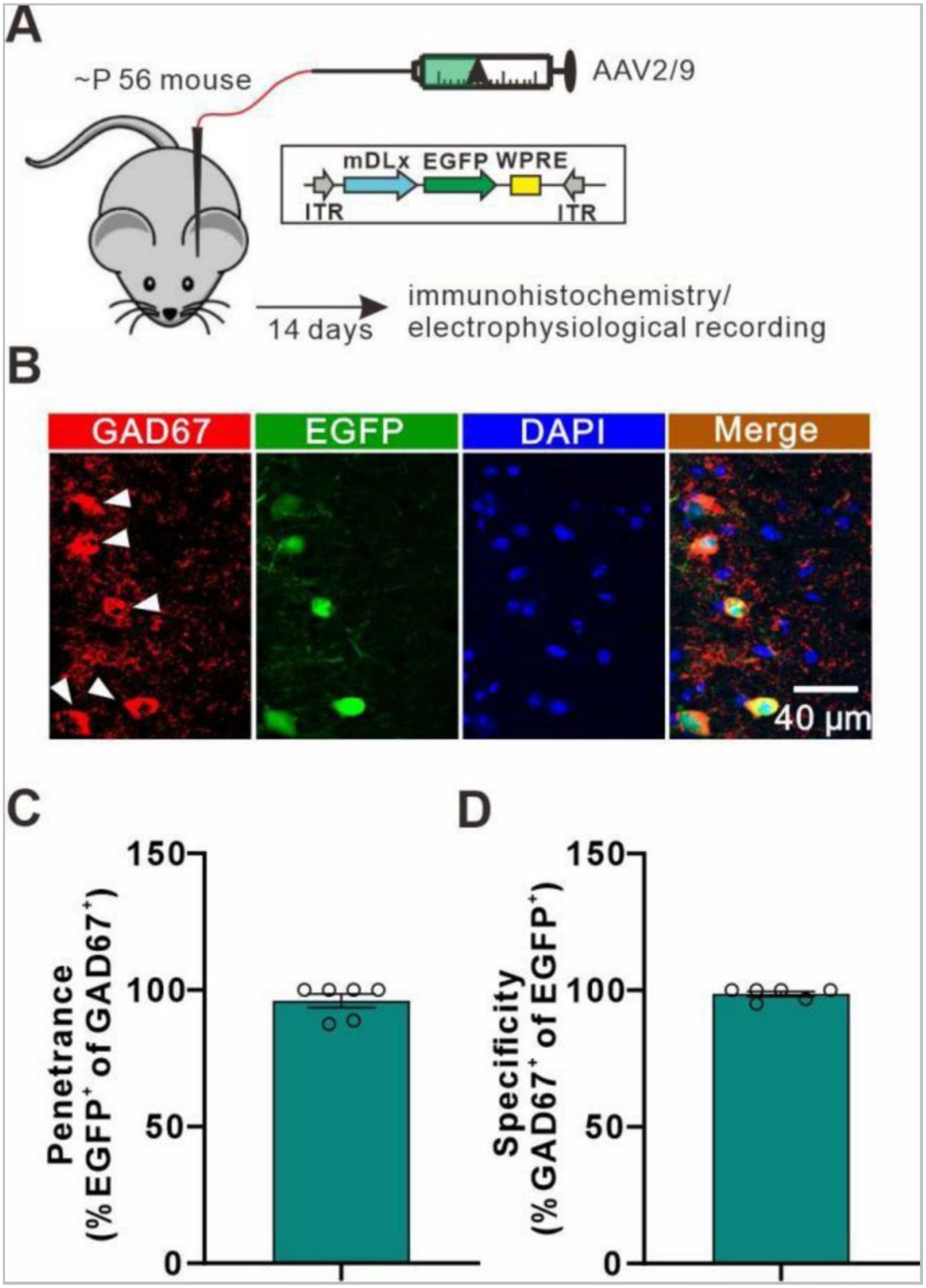

Fig. S1.

EGFP is expressed in GABAergic interneurons in the stratum radiatum of hippocampus.

(A) Schematic illustrating the procedure of AAV2/9 microinjections into the stratum radiatum of the hippocampus.
(B) Confocal images showing EGFP (green) and GAD67 (red, white arrow) expressing cells in the stratum radiatum of the hippocampus.
(C and D) mDLx::EGFP was expressed in >98% of interneurons in stratum radiatum of the hippocampus, with >98% specificity.

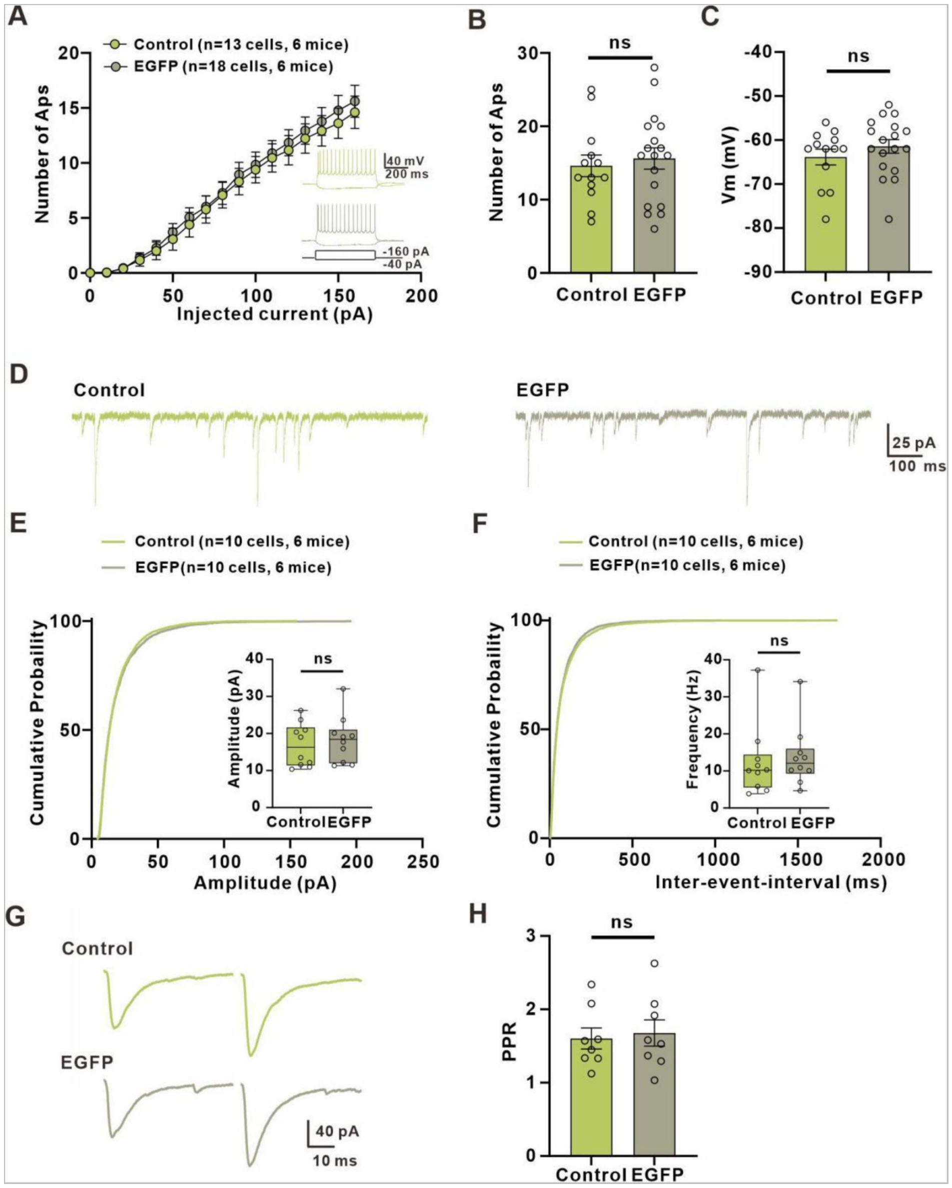

Fig. S2.

EGFP+ interneurons have normal sEPSC, excitability and PPR.

(A) Relationship of injected current to number of APs in postulated interneurons and EGFP+ interneurons. Inset: traces of membrane responses to current injection.

(B) The number of APs at up state membrane potentials (Vm) values **(Control: 14.62±1.474, n=13 slices from 6 mice; EGFP: 15.61±1.458, n=18 slices from 6 mice;, t=0.4684 with 29 degrees of freedom, *p*=0.643, two-tailed unpaired t-test).**

(C) Interneuron resting membrane potentials **(Control: −63.85±1.779 mV, n=13 slices from 6 mice; EGFP: −61.44±1.539 mV, n=18 slices from 6 mice; t=1.018 with 29 degrees of freedom, *p*=0.317, two-tailed unpaired t-test**).

(D) Left: example traces of sEPSCs measured in stratum radiatum of hippocampal postulated interneurons of WT mice; Right: example traces of sEPSCs measured in stratum radiatum of hippocampal EGFP+ interneurons of virus injected mice.

(E, F) Cumulative distribution plots and summary of sEPSC amplitude and frequency in postulated interneurons and EGFP+ interneurons **(Amplitude: 16.89±1.846 pA of control, 18.28±2.003 of EGFP, n=10 slices from 6 mice, t=0.5125 with 18 degrees of freedom, *p*=0.6146, two-tailed unpaired t-test; Frequency: 12.41±3.058 Hz of control, 13.78±2.601 Hz of EGFP, n=10 slices from 6 mice, t=0.3405 with 18 degrees of freedom, *p*=0.7374, two-tailed unpaired t-test).**

(G) Example paired-pulse traces measured in postulated interneurons and EGFP+ interneurons.

(H) Summary of paired-pulse ratio (**Control: 1.602±0.1443, n=8 slices from 3 mice; EGFP: 1.677±0.1794, n=8 slices from 3 mice; t=0.3267 with 14 degrees of freedom, *p*=0.7488, two-tailed unpaired t-test**).

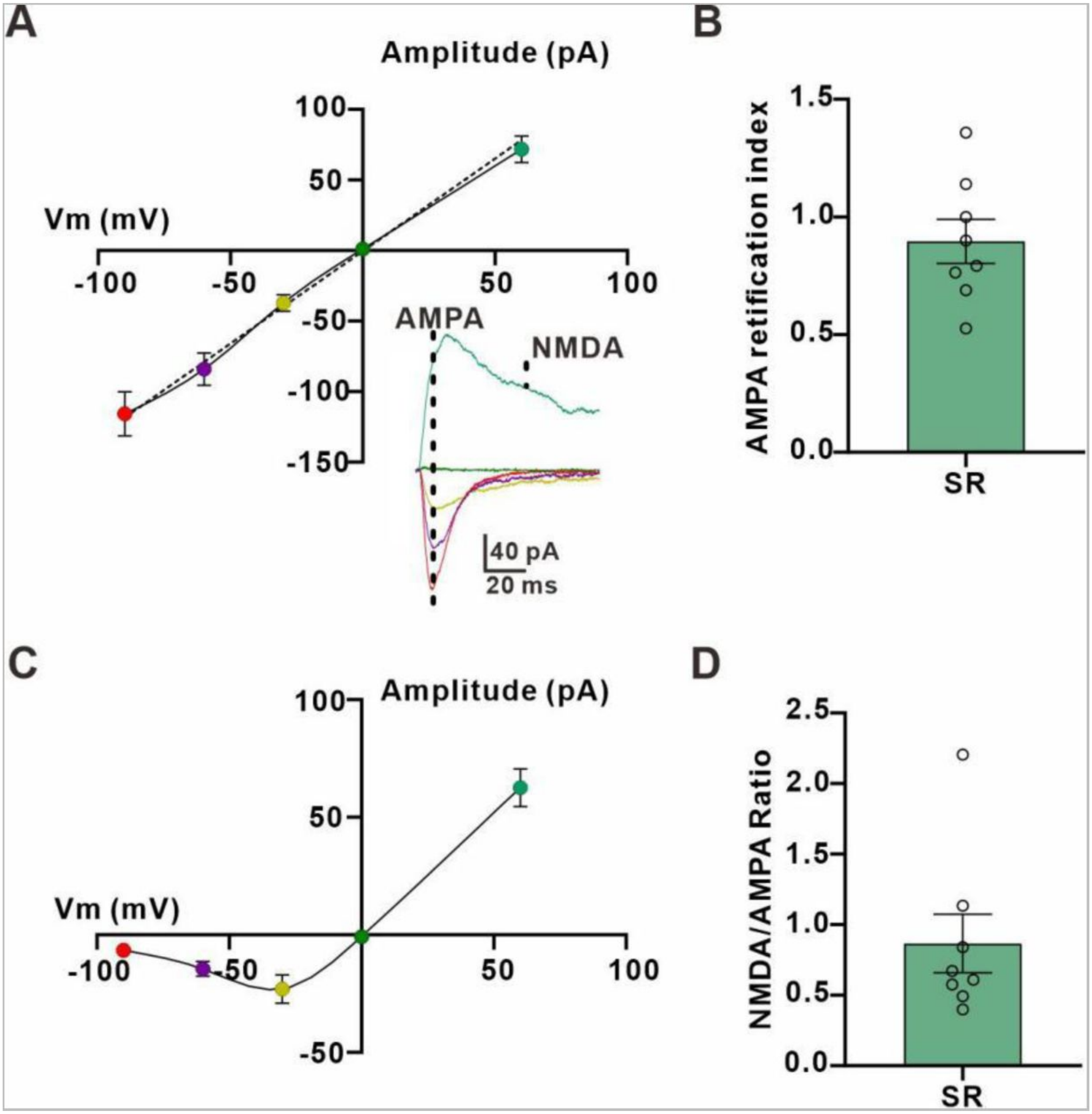

Fig. S3.

Stratum radiatum interneurons show linear rectifying AMPARs and a big NMDAR-mediated component.

(A) Current-voltage (I-V) relation of AMPAR-mediated EPSCs in stratum radiatum interneurons. Inset: averaged EPSCs traces at −90, −60, −30, 0 and 60 mV, showing the times at which the two components were measured.
(B) AMPA rectification index (EPSC amplitude at 60 mV / EPSC amplitude at −60 mV) in eight interneurons from stratum radiatum indicates linear rectifying AMPARs **(AMPA retification index: 0.8959±0.09362)**.
(C) I-V relation for the NMDAR-mediated EPSCs in stratum radiatum interneurons.
(D) NMDA / AMPA ratio (NMDAR-mediated EPSC amplitude at 60 mV / AMPAR-mediated EPSC amplitude at −60 mV) in eight cells from stratum radiatum **(NMDA/AMPA ratio: 0.8664±0.2075)**.

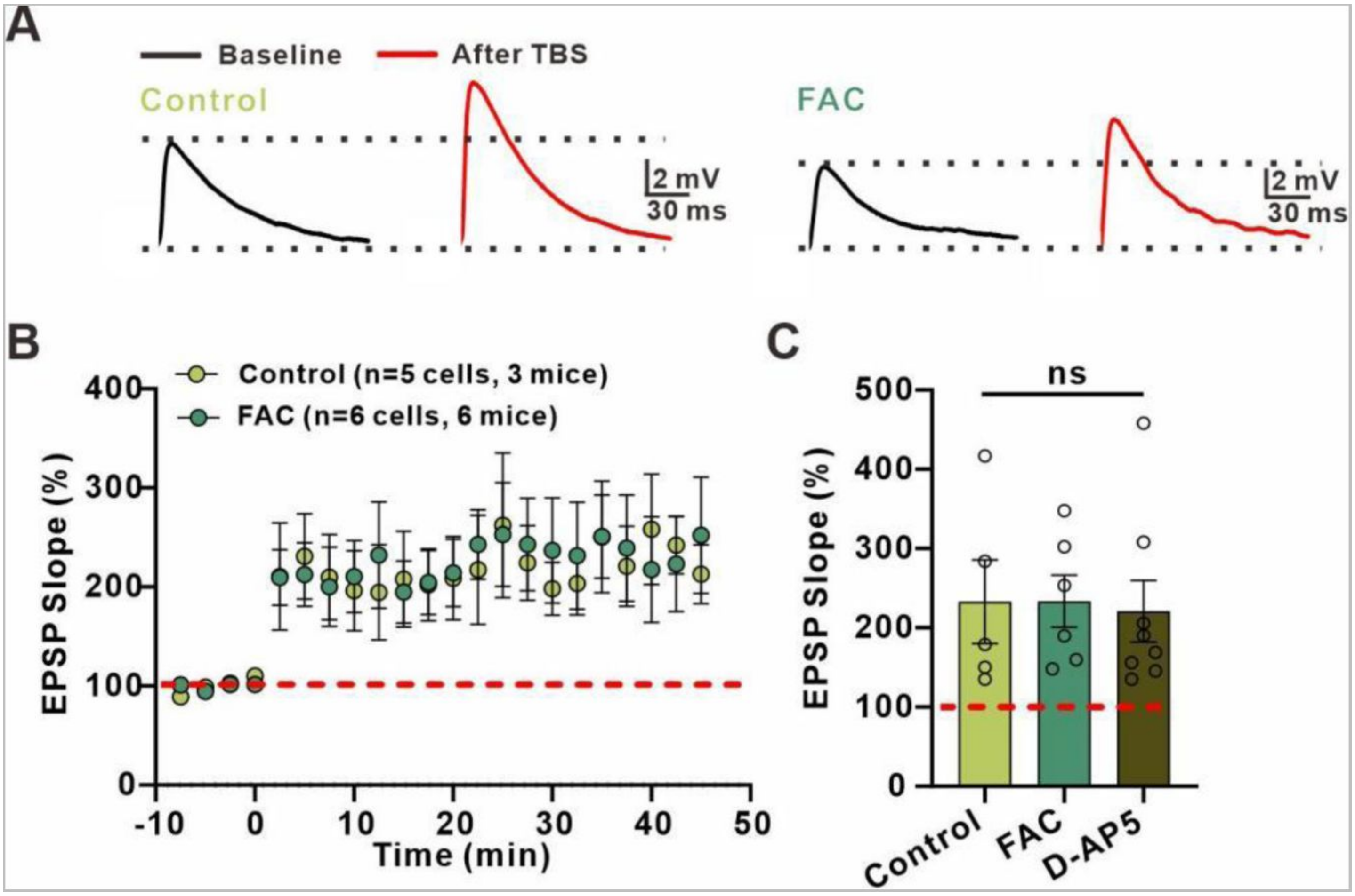

Fig. S4.

$LTP_{E\rightarrow I}$ in stratum oriens is not dependent on astrocytic metabolism and the activation of NMDA receptor.

(A) Left: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) LTP induction; Right: superimposed representative averaged EPSPs recorded 10 min before (dark traces) and 35-45 min after (red traces) LTP induction when astrocytic metabolism was disrupted.

(B) Normalized slope before and after the TBS stimulation protocol in control conditions and in the presence of astrocytic metabolism inhibitor FAC.

(C) The summary data measured 35-45 min after LTP induction **(Control: 233±52.74, n=5 slices from 3 mice, FAC: 233.5±33.06, n=6 slices from 6 mice; D-AP5:220.9±38.94, n=8 slices from 4 mice; *p*=0.96, F(2, 16)=0.03277, ANOVA with Dunnett's comparison**).

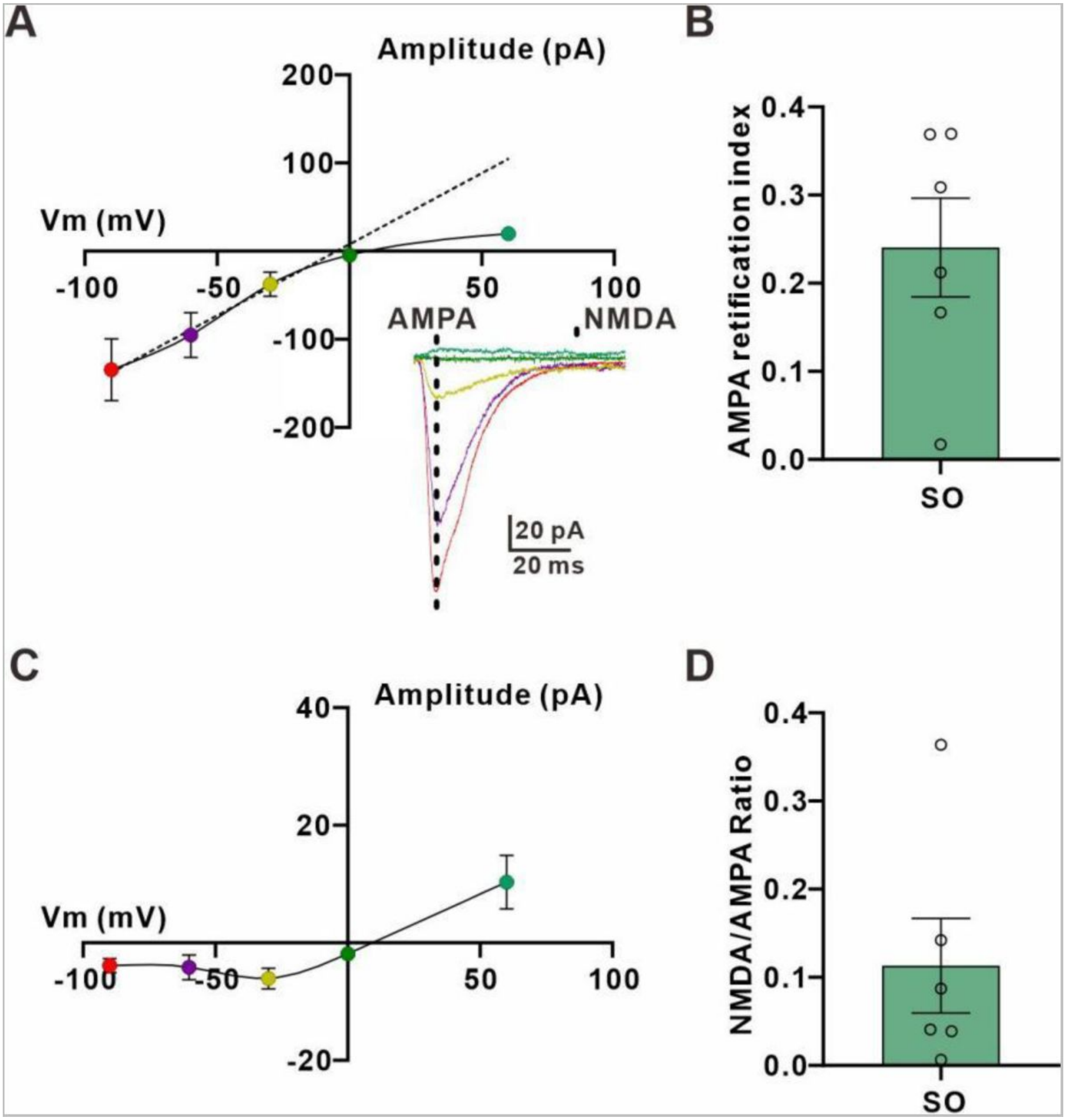

Fig. S5.

Stratum oriens interneurons show inwardly rectifying AMPARs and a negligible NMDAR-mediated component.

(A) Current-voltage (I-V) relation of AMPAR-mediated EPSCs in stratum oriens interneurons. Inset: averaged EPSCs traces at −90, −60, −30, 0 and 60 mV, showing the times at which the two components were measured.

(B) AMPA rectification index (EPSC amplitude at 60 mV / EPSC amplitude at −60 mV) in six interneurons from stratum oriens indicate inwardly rectifying AMPARs **(AMPA retification index: 0.2406±0.05594)**.

(C) I-V relation for the NMDAR-mediated EPSCs in stratum oriens interneurons

(D) NMDA /AMPA ratio (NMDAR-mediated EPSC amplitude at 60 mV / AMPAR-mediated EPSC amplitude at −60 mV) in six cells from stratum oriens **(NMDA/AMPA ratio: 0.1132±0.05370)**.

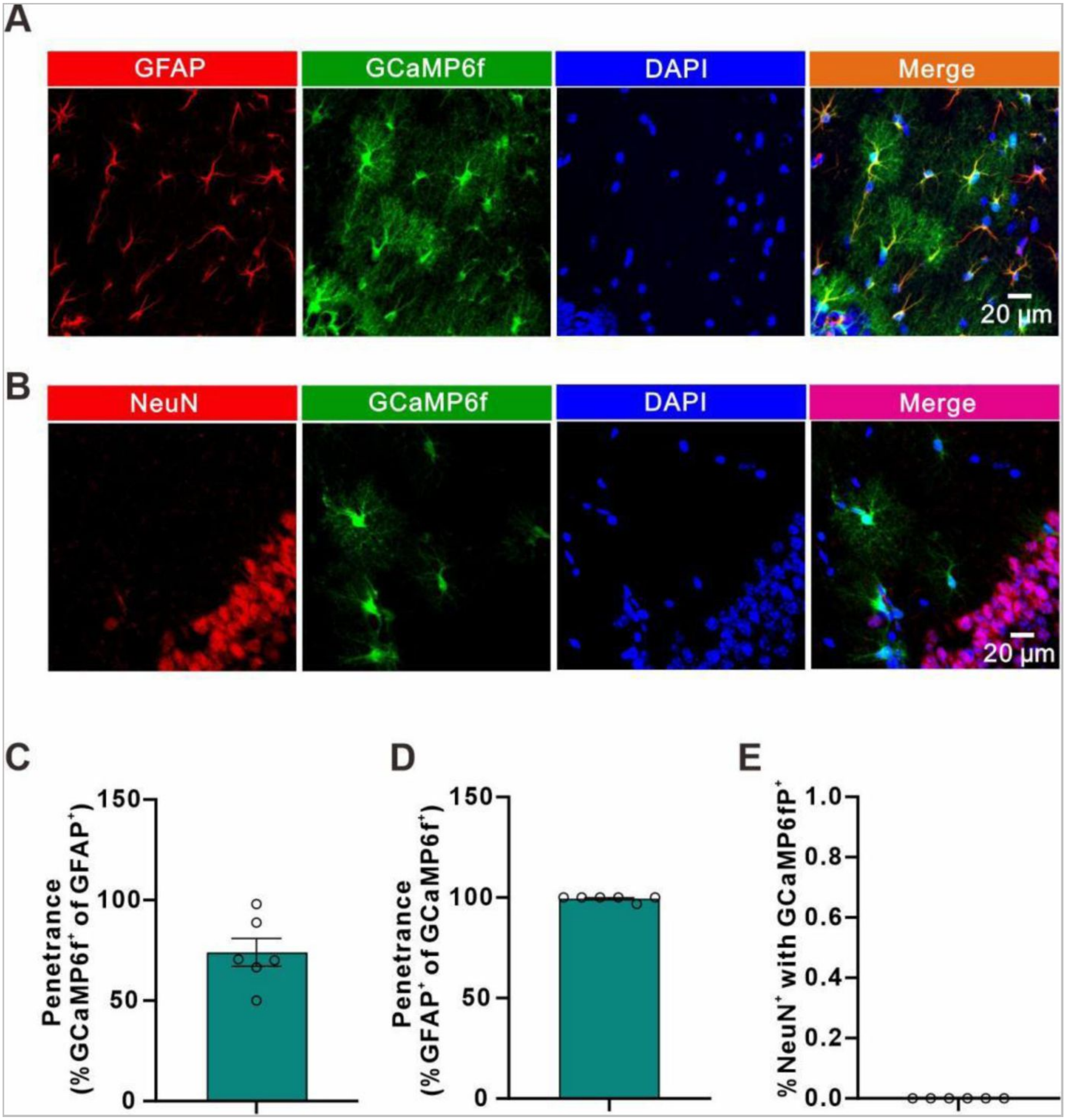

Fig. S6.

GCaMP6f was expressed in astrocytes.

(A and B) Representative immunohistochemistry images showing colocalization of GCaMP6f and GFAP (A), but not NeuN (B).
(C and D) gfaABC$_1$D::GCaMP6f was expressed in >70% of astrocyte in stratum radiatum of the hippocampus, with >99% specificity.
(E) Quantification of the percentage of cells expressing NeuN that also expressed GCaMP6f.

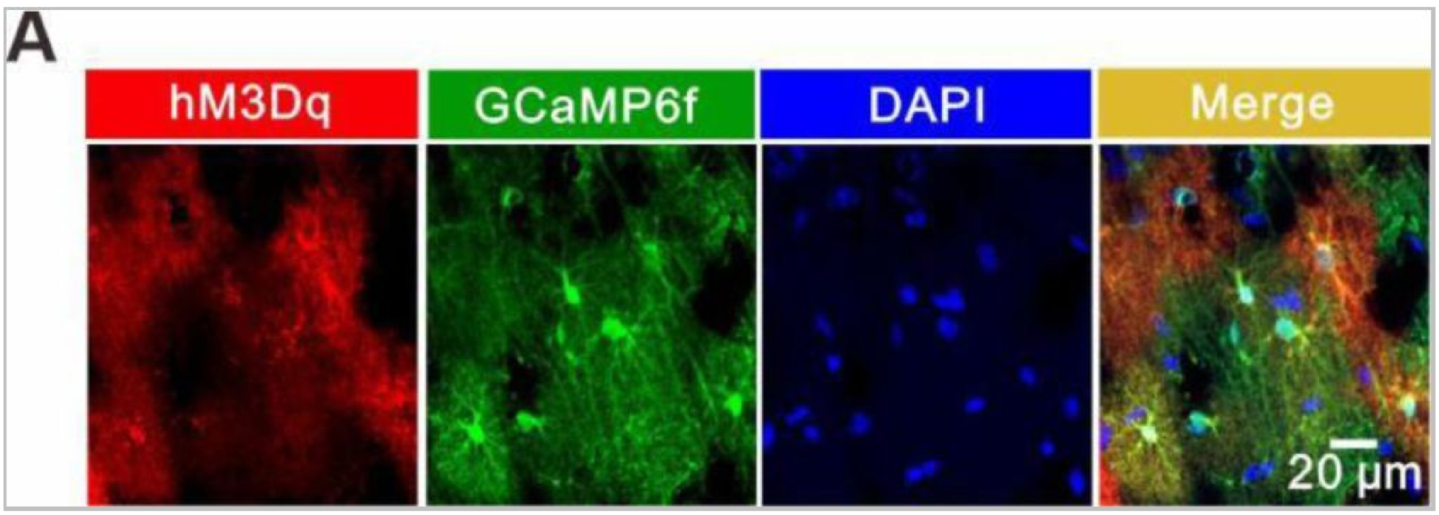

Fig. S7.

hM3Dq was expressed in astrocytes.

(A) hM3Dq(mCherry) colocalization with an GCaMP6f in the stratum radiatum of the hippocampus.

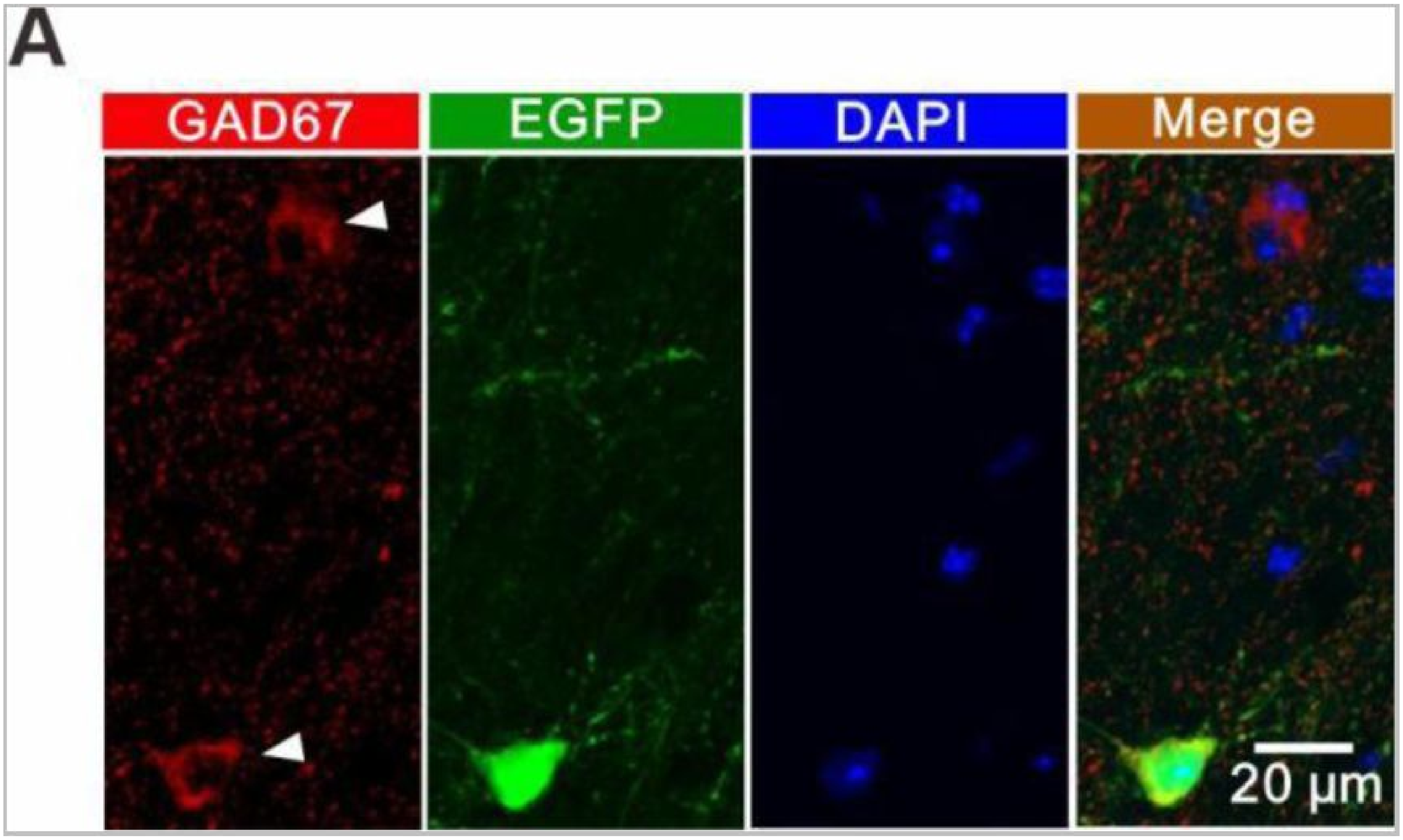

Fig. S8.

EGFP-γCaMKII shRNA is expressed in GABAergic interneurons in the stratum radiatum of hippocampus.

(A) Confocal images showing EGFP γCaMKII shRNA (green) and GAD67 (red, white arrows) expressing cells in the stratum radiatum of the hippocampus. These images indicate that the γCaMKII shRNA, which is driven by an interneuron specific promoter (mDLx), is expressed specifically in interneurons.

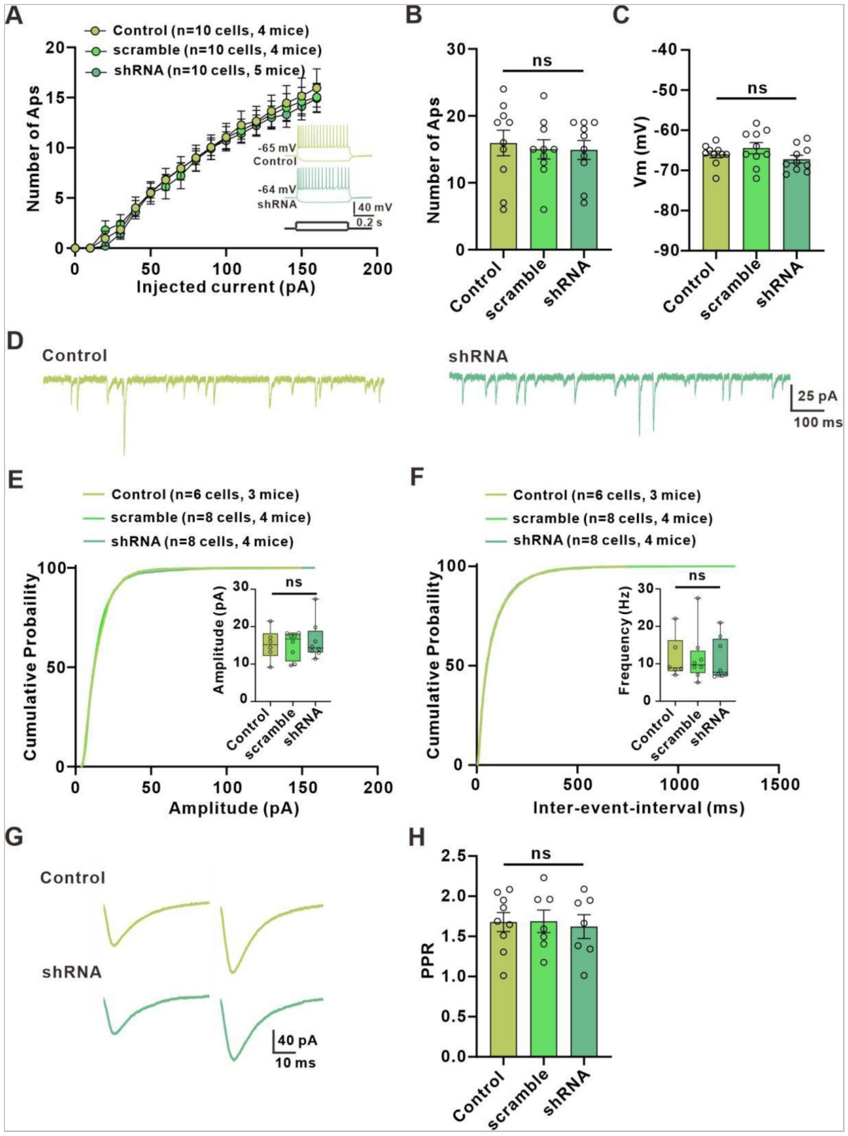

Fig. S9.

knockdown γCaMKII from interneurons have no effect on their sEPSC, excitability and PPR

(A) Relationship of injected current to number of APs in postulated interneurons, expressing scramble shRNA and γCaMKII shRNA interneurons. Inset: traces of membrane responses to current injection.

(B) The number of APs at up state membrane potentials (Vm) values **(Control: 15.95±1.904, n=10 slices from 4 mice; scramble shRNA: 15.00±1.430, n=10 slices from 4 mice; γCaMKII shRNA: 14.90±1.410, n=10 slice from 5 mice; p=0.4986, F(2, 27)=0.4756, ANOVA with Dunnett's comparison)**.

(C) Interneuron resting membrane potentials **(Control: −66.03±0.8127, n=10 slices from 4 mice; scramble shRNA: −64.44±1.398, n=10 slices from 4 mice; γCaMKII shRNA: −67.23±0.9803, n=10 slice from 5 mice; p=0.2128, F(2, 27)=1.639, ANOVA with Dunnett's comparison)**. |

(D) Left: example traces of sEPSCs measured in stratum radiatum of hippocampal postulated interneurons of WT mice; Right: example traces of sEPSCs measured in stratum radiatum of hippocampal EGFP$^+$ interneurons of AAV-mDLx-shRNA γCaMKII expressed mice.

(E, F) Cumulative distribution plots and summary of sEPSC amplitude and frequency in postulated interneurons, expressing scramble shRNA and γCaMKII shRNA interneurons **(Frequency: 11.62±2.328 Hz of control, n=6 slices from 3 mice, 11.69±2.462 Hz of scramble shRNA, n=8 slices from 4 mice, 11.12±2.015 Hz of γCaMKII shRNA, n=8 slices from 4 mice, p=0.9805, F(2, 19)=0.01969, ANOVA with Dunnett's comparison; Amplitude: 15.23±1.846 pA of control, n=6 slices from 3 mice, 14.57±1.382 of scramble shRNA, n=8 slices from 4 mice,16.23±1.821 of γCaMKII shRNA, n=8 slices from 4 mice, p=0.8459, F(2, 19)=0.1688, ANOVA with Dunnett's comparison)**.

(G) Example paired-pulse traces measured in postulated interneurons of WT mice and EGFP$^+$ interneurons of AAV-mDLx-shRNA γCaMKII expressed mice.

(H) Summary of paired-pulse ratio **(Control: 1.679±0.1190 n=9 slices from 3 mice; scramble shRNA: 1.688±0.1405, n=7 slices from 3 mice; γCaMKII shRNA: 1.622±0.1477, n=7 slice from 3 mice; p=0.9361, F(2, 20)=0.06628, ANOVA with Dunnett's comparison)**.

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Reviewer #1 (Public Review):

Summary:
The question at hand is whether astrocytes contribute to the mechanism of long-term synaptic potentiation (LTP) at synaptic contacts between excitatory glutamatergic neurons and inhibitory neurons (E-I synapses). This is a legitimate query considering the immense body of work that has now established synaptic plasticity (LTP, LTD and spike-timing

dependent plasticity) as an astrocyte-dependent process at excitatory synapses and, by contrast, the lack of knowledge on whether and how astrocytes control IN activity. Taking direct inspiration from that same body of work, authors recapitulate a number of experiments and approaches from prior seminal studies and provide evidence that E-I synapses in the stratum radiatum of the hippocampus display NMDAR-dependent plasticity, which can be suppressed by pharmacologically hindering astrocytes physiology, preventing astrocyte Ca^{2+} transients or blocking endocannabinoid CB1 receptors. Under any of these conditions, LTP can still be rescued by exogenously applying D-serine, a naturally occurring co-agonist of NMDARs primarily released by astrocytes. Coincidently, authors show that the conditions used to elicit LTP also cause a transient increase in NMDAR co-agonist site occupancy. Lastly, based on some evidence that gamma-CaMKII is predominantly expressed in INs rather than excitatory neurons, authors conducted AAV-mediated IN-specific gamma-CaMKII shRNA experiments and found that this is sufficient to suppress LTP at E-I synapses. They found that this approach also impairs contextual fear learning in behaving mice. Authors conclude that astrocytes gate LTP at E-I synapses via a mechanism wherein neuronal depolarization during LTP induction elicits endocannabinoid release which drives CB1-dependent astrocyte Ca^{2+} activity, causing the release of the NMDAR co-agonist D-serine (required for NMDAR activation).

Strengths:
This is an important question and the experimental work seems to have been conducted at high standards. The electrophysiology traces are impeccable, the experiments are well powered, including the behavioral testing, and multiple controls and validations are provided throughout. The figures are clear and easy to understand. Overall, the conclusions from the study are consistent, or partially consistent, by the findings.

Main Weaknesses:
1- A major point of concern is the lack of proper acknowledgment of the seminal studies that were mimicked in this manuscript, notably Henneberger et al, Nature 2010, Adamsky et al, Cell 2018; and Robin et al., Neuron 2017. The entire study design is a replica of these landmark studies: it isn't built upon or inspired from them, it exactly repeats the experiments and methods performed in them, coming dangerously close to being simply a hidden attempt to plagiarize published work. The resemblance goes as far as using an identical figure display (see Fig4.D vs Fig 2D of Ref#4). The issue is that authors frame the problem, scientist logic, reasoning, technical tricks, approaches, and interpretations as their own whereas, in reality, they were taken verbatim out of previous work and applied to a (shockingly) similar problem. The probity of the present study is thus in question. Authors need to clearly acknowledge, in all relevant instances, that the work presented here recapitulates the approach, reasoning and methodology used in past seminal studies that tackled the mechanisms of astrocyte regulation of LTP.

2-Relatedly, in past work, field recordings were used to monitor LTP in hippocampal slices (refs 4, 26 and others). This method captures indiscriminately all excitatory synapses where glutamate is released to cause AMPAR-dependent (and NMDAR) transmembrane flux of cations in the postsynaptic element, including E-I synapses and not just E-E synapse like the authors claim. Therefore, a strong argument can be made that there is no actual ground to differentiate the present results from past ones.

3-There is a general lack of excitement about this study. One reason is that it replicates almost identically past work, as mentioned above. Another is that the scientific question and importance of the findings are not framed appropriately. The work is presented as an astrocyte-focused investigation, but it has very limited value to the astrocyte field. The findings are, in all accounts, identical to those unveiled by previous work especially because E-I synapses are, in fact, excitatory synapses. Where this study does bring value, however, is to the field of interneurons, but it would need to be reframed to shift the emphasis from astrocytes to E-I connections. Authors would need to elevate the text by framing their work around relevant considerations, such as IN diversity, mechanisms of LTP in IN subtypes, role of E-I connections in hippocampal circuit function, information processing, behavior, spatial learning, navigation, or grid cells activity etc...

4-A clear weakness of the study is that it fails to consider the molecular and functional diversity of interneurons in the stratum radiatum and provides no insights or considerations related to it. Authors provide no information on what type of IN were patched, or the location of their cell body in the s.r., effectively treating all patched IN as a homogeneous ensemble of cells - which they are not. Relatedly, the study is extremely evasive on the importance of the results in the context of inhibitory interneurons. This renders the significance of the insights highly uncertain and dampens both the impact of the study and the excitement it generates. Hippocampal interneurons are very diverse in molecular identity, sub-anatomical location, morphology, projections, connectivity and functional importance. Some experts go as far as recognizing 29 subtypes in the CA1, including 9 in the stratum radiatum alone (based on the location of their soma). However, this is neither addressed nor acknowledged by the authors, with the exception of a statement (line 659) where they claim to have "focused on a subpopulation of interneurons in the stratum radiatum" without providing any precision or evidence to corroborate this assertion. This diversity, alone, could explain why not all cells showed LTP, or why the mechanisms authors describe in the radiatum do not seem to be at play in the oriens. Hence, carefully considering the diversity of INs in the present work is necessary. It would refine and augment the conclusions of the paper. Instead of a sub-region specificity, the study might fuel the notion of an IN subtype specificity of LTP mechanisms, which is more useful to the field.

5-Authors take several shortcuts. Some of the conclusions are a leap from the experiments and are only acceptable due to the close analogy with very similar investigations conducted in the past that provided identical results. For instance, the present study provides no evidence of any sort that D-serine is involved - rather, it provides evidence that the pathway at hand contributes to increasing the occupancy of the co-agonist binding site of NMDARs. Considering the absence of work demonstrating that D-serine is the endogenous co-agonist of NMDARs at E-I synapses, most of the authors claims on D-serine are unfounded. This would necessitate using tools such as the canonical D-serine scavengers DAAS or DsDA, serine racemase KO mice etc. Similarly, authors provide no compelling evidence that endocannabinoid CB1 receptors involved in this pathway are located on astrocytes

6-An important caveat in this study is the protocol employed to induce LTP, which includes steps of sustained depolarization of the patched IN to -10mV. Neuronal depolarization is known to induce endocannabinoids production. In several instances, this was shown to 'activate' astrocytes and elicit the release of astrocyte-derived transmitters at nearby synapses. This implies that the endocannabinoid-dependent pathway described in the study is, most likely, artificially engaged by the protocol itself. Hence, the present work only provides evidence that an astrocyte-dependent, CB1-D-serine-pathway can be artificially called upon with this specific LTP protocol, but does not convincingly demonstrate that it is naturally occurring or necessary for plasticity at E-I synapses. Authors would need to thoroughly address this caveat by replicating some of their key findings (AM251, calcium-clamp, D-serine and CaMKII shRNA) using a protocol that does not entail the artificial depolarization of the patched interneuron.

7-Reading and understanding are hindered by a rather vast array of issues with the text itself. It needs thorough editing for typos, misnomers, meaning-altering errors in syntax, and a number of issues with English.

Reviewer #2 (Public Review):

Summary:
This work explores the implication of astrocytes in the regulation of long-term potentiation of excitatory synapses onto inhibitory neurons in CA1 hippocampus. They found that astrocytes of a sub-region of CA1 regulate this plasticity through their activation of endocannabinoids that lead to the release of the NMDA receptor co-agonist, D-serine.

Strengths:
The experiments are well considered and conceptualized, and use appropriate tools to explore the role of astrocytes in the tripartite synapse. The results highlight a novel role of astrocytes in an important aspect of the synaptic regulation of the hippocampal circuit. There are extensive levels of analysis for each experimental group of evidence.

Weaknesses:
The authors underscore and used an oversimplified view of the heterogeneity of interneuron populations and their selective roles in the hippocampal network. Also, there is an uneven level of astrocyte-selective tools used in the different experiments which creates an uneven strength of arguments and conclusions regarding the role of glial cells. Finally, the wording used by the authors often lead to some confusion or sense of overinterpretation.

Author Response

We are very thankful for the editors' and reviewers' thoughtful feedback and criticisms on our manuscript. We have carefully considered all of the comments and will provide a revised manuscript with detailed responses as soon as we can. In the meantime, we will make our best effort to conduct additional experiments to further support our conclusions.We greatly appreciate the time and consideration given to improving our work.

> ### *Reviewer #1 (Public Review):*
>
> *Summary:*
>
> *The question at hand is whether astrocytes contribute to the mechanism of long-term synaptic potentiation (LTP) at synaptic contacts between excitatory glutamatergic neurons and inhibitory neurons (E-I synapses). This is a legitimate query considering the immense body of work that has now established synaptic plasticity (LTP, LTD and spike-timing dependent plasticity) as an astrocyte-dependent process at excitatory synapses and, by contrast, the lack of knowledge on whether and how astrocytes control IN activity. Taking direct inspiration from that same body of work, authors recapitulate a number of experiments and approaches from prior seminal studies and provide evidence that E-I synapses in the stratum radiatum of the hippocampus display NMDAR-dependent plasticity, which can be suppressed by pharmacologically hindering astrocytes physiology, preventing astrocyte Ca2+ transients or blocking endocannabinoid CB1 receptors. Under any of these conditions, LTP can still be rescued by exogenously applying D-serine, a naturally occurring co-agonist of NMDARs primarily released by astrocytes. Coincidently, authors show that the conditions used to elicit LTP also cause a transient increase in NMDAR co-agonist site occupancy. Lastly, based on some evidence that gamma-CaMKII is predominantly expressed in INs rather than excitatory neurons, authors conducted AAV-mediated IN-specific gamma-CaMKII shRNA experiments and found that this is sufficient to suppress LTP at E-I synapses. They found that this approach also impairs contextual fear learning in behaving mice. Authors conclude that astrocytes gate LTP at E-I synapses via a mechanism wherein neuronal depolarization*

> *during LTP induction elicits endocannabinoid release which drives CB1-dependent astrocyte Ca2+ activity, causing the release of the NMDAR co-agonist D-serine (required for NMDAR activation).*
>
> *Strengths:*
>
> *This is an important question and the experimental work seems to have been conducted at high standards. The electrophysiology traces are impeccable, the experiments are well powered, including the behavioral testing, and multiple controls and validations are provided throughout. The figures are clear and easy to understand. Overall, the conclusions from the study are consistent, or partially consistent, by the findings.*

We greatly appreciate you taking the time to evaluate our study thoroughly and provide such thoughtful feedback.

> *Main Weaknesses:*
>
> 1. *A major point of concern is the lack of proper acknowledgment of the seminal studies that were mimicked in this manuscript, notably Henneberger et al, Nature 2010, Adamsky et al, Cell 2018; and Robin et al., Neuron 2017. The entire study design is a replica of these landmark studies: it isn't built upon or inspired from them, it exactly repeats the experiments and methods performed in them, coming dangerously close to being simply a hidden attempt to plagiarize published work. The resemblance goes as far as using an identical figure display (see Fig4.D vs Fig 2D of Ref#4). The issue is that authors frame the problem, scientist logic, reasoning, technical tricks, approaches, and interpretations as their own whereas, in reality, they were taken verbatim out of previous work and applied to a (shockingly) similar problem. The probity of the present study is thus in question. Authors need to clearly acknowledge, in all relevant instances, that the work presented here recapitulates the approach, reasoning and methodology used in past seminal studies that tackled the mechanisms of astrocyte regulation of LTP.*

Thank you very much for your review and valuable comments on our manuscript. We greatly appreciate your concern regarding the proper acknowledgment of previous studies. We sincerely apologize for not adequately citing and acknowledging the seminal works in our manuscript. We highly value avoiding academic misconduct.

For the research design, although there are some similarities between our work and other studies, our key scientific questions and technical approaches are markedly different, as evidenced by our central hypothesis and experimental methods. We did not completely replicate their research design.

Regarding research methods, many basic techniques like electrophysiology, chemogenetic are common experimental methods, not patented by any one paper. Our choice of methods is based on the research needs, not to replicate a particular paper. But we recognize that there are similarities in our experimental methods, specifically the chemogenetic stimulation of astrocytes to induce de novo LTP, which has been inspired by previous studies (Van Den Herrewegen et al. Molecular Brain (2021), Adamsky et al. Cell (2018), Nam et al. Cell reports (2019)). We were also inspired by the previous work of Henneberger et al. in Nature (2010) to investigate whether stimulation, specifically we using TBS (theta burst stimulation), could transiently increase NMDA receptor-mediated synaptic responses.

For the similarity between our Fig. 4D and Fig. 2D of Ref. 4, it is primarily because both studies have the similar purpose(we monitored NMDA currents in interneurons, others monitored in pyramidal cells) using similar methods, but our figure layout follows a regular display pattern. Additionally, we would like to draw your attention to our previous studies, specifically Shen et al., Scientific Reports (2017), Supplementary figure 4, and Shen et al., Journal of Neurochemistry (2021), Supplementary figures 8 and 9. In these studies, we also employed a regular display pattern in our figure layouts. It is important to note that while there may be similarities in the figure arrangement, each study presents distinct findings and contributes to the broader understanding of the topic.Our use of a similar way to present data does not equal plagiarism. We apologize for any confusion caused by the lack of explicit citation and acknowledgment in our manuscript again. In the revised version, we will ensure to provide clear and detailed references to all relevant studies.

In terms of citations, we have cited Henneberger et al, Nature 2010, Adamsky et al, Cell 2018; and Robin et al., Neuron 2017.'s work in multiple places, indicating we have learned from their research ideas and findings. We will supplement any missing citations. But overall, our work has distinct differences and innovations.

We are not intended as a hidden attempt to plagiarize or simply replicate their methods. Rather, they are part of a deliberate effort to establish a comparable and reproducible experimental framework. Our study aims to validate and further explore the conclusions drawn by replicating the experiments of these seminal studies and deepening our understanding of the mechanisms of astrocyte regulation of LTPE-I.

We sincerely appreciate your review and guidance. We will carefully consider your criticism and incorporate more accurate and thorough citations in the revised version, ensuring proper respect and acknowledgment of the previous works.

> 1. *Relatedly, in past work, field recordings were used to monitor LTP in hippocampal slices (refs 4, 26 and others). This method captures indiscriminately all excitatory synapses where glutamate is released to cause AMPAR-dependent (and NMDAR) transmembrane flux of cations in the postsynaptic element, including E-I synapses and not just E-E synapse like the authors claim. Therefore, a strong argument can be made that there is no actual ground to differentiate the present results from past ones.*

Thank you for your thoughtful comments regarding the differentiation of our results from previous studies. We appreciate the opportunity to address this issue and provide further clarification.

Indeed, in past studies, field recordings were commonly utilized to monitor long-term potentiation (LTP) in hippocampal slices. It is true that this method captures all flux of cations in excitatory synapses, inhibitory synapses and glia. This includes both excitatory-excitatory (E-E) and excitatory-inhibitory (E-I) synapses.

When using the LTP recording protocol, one limitation is that the experimenter cannot determine the exact contribution of E-E and E-I currents to the recorded current. Additionally, it is not possible to know, with the same induction protocol, the specific effects on E-E synapses versus E-I synapses. It is plausible that E-E synapses could undergo LTP, while E-I synapses could undergo LTD, or vice versa.

Thus, it becomes crucial to carefully dissect the functioning of E-I synapses and investigate how astrocytes modulate these synapses. Past field recordings have provided important insights, our selective interrogation of the astrocyte-E-I synapse interface represents a conceptual advance to delineate the nuanced modulation of distinct synaptic connections by astrocytes. We specifically focus on studying the modulation of E-I synapses by astrocytes and aim to elucidate the intricate dynamics and underlying mechanisms. By untangling the complex contributions of astrocytes to E-I synapse function and plasticity, we can unveil novel aspects of neuroglial interactions and advance our understanding of the fundamental principles governing neural network activity.

> 1. *There is a general lack of excitement about this study. One reason is that it replicates almost identically past work, as mentioned above. Another is that the scientific question and importance of the findings are not framed appropriately. The work is presented as an astrocyte-focused investigation, but it has very limited value to the astrocyte field. The findings are, in all accounts, identical to those unveiled by previous work especially because E-I synapses are, in fact, excitatory synapses. Where this study does bring value, however, is to the field of interneurons, but it would need to be reframed to shift the emphasis from astrocytes to E-I connections. Authors would need to elevate the text by framing their work around relevant considerations, such as IN diversity, mechanisms of LTP in IN subtypes, role of E-I connections in hippocampal circuit function, information processing, behavior, spatial learning, navigation, or grid cells activity etc...*

We appreciate your insightful comments and concerns regarding the lack of excitement surrounding our study. We would like to clarify that while our study use similar certain methodologies, for example electrophysiology, chemogenetics and pharmacology, our research aims to provide a deeper understanding of the underlying mechanisms of how astrocytes regulate E-I synapses. We apologize if this replication aspect was not adequately highlighted in our manuscript, and we will make sure to emphasize the novel contributions of our study in the revised version.

Regarding the framing of our study, we recognize the importance of interneurons and the role of E-I connections in hippocampal circuit function, information processing, behavior, spatial learning, navigation, and other relevant aspects. However, the scientific question and scope of the study are to explore whether and how astrocytes modulate E-I synapses. We believe that this study brings value to the field of astrocyte-neuron interaction. Of course, this study also brings value to the field of interneurons. Perhaps the lack of excitement among audiences stems from the mechanisms for astrocytes modulating E-I and E-E synapses are the same.

> 1. *A clear weakness of the study is that it fails to consider the molecular and functional diversity of interneurons in the stratum radiatum and provides no insights or considerations related to it. Authors provide no information on what type of IN were patched, or the location of their cell body in the s.r., effectively treating all patched IN as a homogeneous ensemble of cells - which they are not. Relatedly, the study is extremely evasive on the importance of the results in the context of inhibitory interneurons. This renders the significance of the insights highly uncertain and dampens both the impact of the study and the excitement it generates. Hippocampal interneurons are very diverse in molecular identity, sub-anatomical location, morphology, projections, connectivity and functional importance. Some experts go as far as recognizing 29 subtypes in the CA1, including 9 in the stratum radiatum alone (based on the location of their soma). However, this is neither addressed nor acknowledged by the authors, with the exception of a statement (line 659) where they claim to have "focused on a subpopulation of interneurons in the stratum radiatum" without providing any precision or evidence to corroborate this assertion. This diversity, alone, could explain why not all cells showed LTP, or why the mechanisms authors describe in the radiatum do not seem to be at play in the oriens. Hence, carefully considering the diversity of INs in the present work is necessary. It would refine and augment the conclusions of the paper. Instead of a sub-region specificity, the study might fuel the notion of an IN subtype specificity of LTP mechanisms, which is more useful to the field.*

Thank you very much for your review and valuable comments on our study. We agree with the point you raised regarding a clear weakness in our study, specifically the lack of consideration the diversity of interneurons in the stratum radiatum.

As the reviewer notes, there are many subtypes of interneurons in hippocampal region CA1 that likely contribute in distinct ways to circuit function. Unfortunately we did not gather information on the specific molecular or morphological identity of the interneurons we recorded from.This is a limitation of our study. We will add discussion of this issue as a caveat, and highlighted it as an opportunity for future work to dissect how long-term potentiation in interneurons regulated by astrocytes may differ across interneuron subpopulations. Thank you once again for your insightful comments.

> 1. *Authors take several shortcuts. Some of the conclusions are a leap from the experiments and are only acceptable due to the close analogy with very similar investigations conducted in the past that provided identical results. For instance, the present study provides no evidence of any sort that D-serine is involved - rather, it provides evidence that the pathway at hand contributes to increasing the occupancy of the co-agonist binding site of NMDARs. Considering the absence of work demonstrating that D-serine is the endogenous co-agonist of NMDARs at E-I synapses, most of the authors claims on D-serine are unfounded. This would necessitate using tools such as the canonical D-serine scavengers DAAS or DsDA, serine racemase KO mice etc. Similarly, authors provide no compelling evidence that endocannabinoid CB1 receptors involved in this pathway are located on astrocytes*

Thank you for your insightful comments on our study. We appreciate your attention to detail and your concerns regarding our conclusions. We agree that further evidence is needed to establish the involvement of D-serine as the endogenous co-agonist of NMDARs at E-I synapses. We will take into consideration your suggestion of using tools such as D-serine scavengers to provide clearer evidence.

Regarding the involvement of endocannabinoid CB1 receptors on astrocytes in this pathway, we provide evidence that astrocytic calcium signaling could blocked by CB1 receptor antagonist AM251, as shown in figure 3.However, we agree that further research is necessary to accurately identify the localization of CB1 receptors. As part of our future investigations, we will take note of this limitation in our discussion and emphasize the need for additional studies to explore the precise location of CB1 receptors. In addition, we will endeavor to perform immunohistochemistry to identify the exact location of CB1 receptors in astrocytes.

Thank you once again for your valuable feedback. We will carefully address these concerns and make appropriate revisions to ensure the clarity and accuracy of our findings.

> 1. *An important caveat in this study is the protocol employed to induce LTP, which includes steps of sustained depolarization of the patched IN to -10mV. Neuronal depolarization is known to induce endocannabinoids production. In several instances, this was shown to 'activate' astrocytes and elicit the release of astrocyte-derived transmitters at nearby synapses. This implies that the endocannabinoid-dependent pathway described in the study is, most likely, artificially engaged by the protocol itself. Hence, the present work only provides evidence that an astrocyte-dependent, CB1-D-serine-pathway can be artificially called upon with this specific LTP protocol, but does not convincingly demonstrate that it is naturally occurring or necessary for plasticity at E-I synapses. Authors would need to thoroughly address this caveat by replicating some of their key findings (AM251, calcium-clamp, D-serine and CaMKII shRNA) using a protocol that does not entail the artificial depolarization of the patched interneuron.*

Thank you for raising this important point. We agree that the sustained depolarization protocol we used to induce LTP could potentially engage endocannabinoid signaling and astrocyte activation. However, we observed that preventing astrocyte Ca^{2+} transients or blocking endocannabinoid CB1 receptors prevented the induction of LTP by this depolarization protocol suggests that this astrocyte-endocannabinoid-dependent pathway is necessary,

Importantly, synaptic depolarization of neurons can occur naturally during learning and memory. Though 'artificial' here, our protocol may mimic aspects of natural activity patterns that engage 'endocannabinoid release' and astrocyte involvement in plasticity.

Another limitation of our study is that we currently cannot conclusively determine the source of the CB1. We cannot distinguish whether the CB1 originates from neurons or astrocytes based on our current experiments. We will explicitly acknowledge this caveat in the discussion, noting that further experiments are needed to clarify the cellular origin of the CB1. Thank you for drawing our attention to this critical issue - we will refine the manuscript accordingly to more comprehensively and accurately present the study conclusions and limitations. Your feedback helps improve the rigor of our research.

> 1. *Reading and understanding are hindered by a rather vast array of issues with the text itself. It needs thorough editing for typos, misnomers, meaning-altering errors in syntax, and a number of issues with English.*

Thank you very much for your review and feedback on our text. We highly appreciate your comments and take them seriously. We will carefully address the issues you mentioned and thoroughly edit the text to eliminate any typos, misnomers, syntax errors that may alter the meaning, and other English-related issues. We truly value your input and appreciate your patience as we work on these improvements.

> ## *Reviewer #2 (Public Review):*
>
> *Summary:*
>
> *This work explores the implication of astrocytes in the regulation of long-term potentiation of excitatory synapses onto inhibitory neurons in CA1 hippocampus. They found that astrocytes of a sub-region of CA1 regulate this plasticity through their activation of endocannabinoids that lead to the release of the NMDA receptor co-agonist, D-serine.*
>
> *Strengths:*
>
> *The experiments are well considered and conceptualized, and use appropriate tools to explore the role of astrocytes in the tripartite synapse. The results highlight a novel role of astrocytes in an important aspect of the synaptic regulation of the hippocampal circuit. There are extensive levels of analysis for each experimental group of evidence.*

Thank you for your positive feedback on our study. We appreciate your recognition of the careful consideration and conceptualization of our experiments, as well as the use of appropriate tools to investigate the role of astrocytes in the tripartite synapse. We are pleased to hear that the results have highlighted a novel role of astrocytes in an important aspect of synaptic regulation in the hippocampal circuit.

Thank you for taking the time to review our work and for providing such positive feedback. We will continue to improve and refine our study based on your valuable comments.

> *Weaknesses:*
>
> *The authors underscore and used an oversimplified view of the heterogeneity of interneuron populations and their selective roles in the hippocampal network. Also, there is an uneven level of astrocyte-selective tools used in the different experiments which*

> *creates an uneven strength of arguments and conclusions regarding the role of glial cells. Finally, the wording used by the authors often lead to some confusion or sense of overinterpretation*

We appreciate the reviewer raising these important points about the characterization of interneuron and astrocyte populations in our study. We agree that oversimplifying or overlooking cellular heterogeneity could undermine the conclusions. In the revised manuscript, we will:

1. Add more detailed discussion of interneuron diversity. We will note this as an area for further study.

2. Review the wording used when describing results and conclusions, ensuring we avoid overstating interpretations of the data.

Thank you again for the thoughtful feedback.